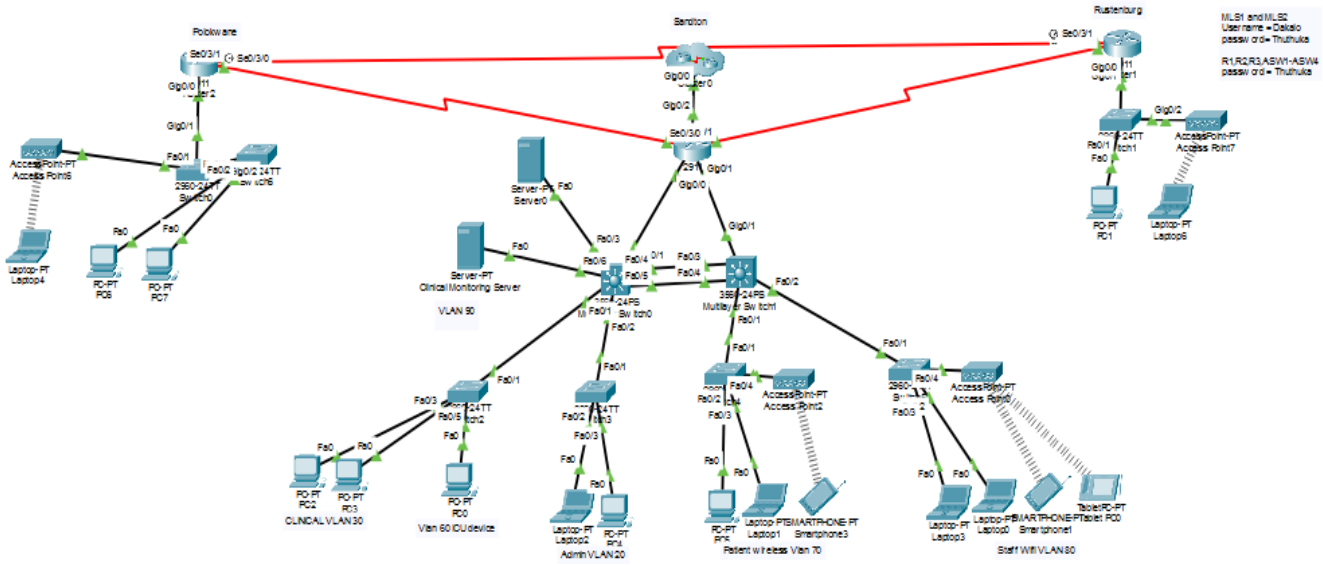
Bradley Dakalo Mvula

All passwords in my packet tracer devices

Device	Username	password
MLS1	Dakalo	Thuthuka
MLS2	Dakalo	Thuthuka
R1	-	Thuthuka
R2	-	Thuthuka
R3	-	Thuthuka
ASW1	-	Thuthuka
ASW2	-	Thuthuka
ASW3	-	Thuthuka
ASW4	-	Thuthuka
PSW1&2	-	Thuthuka
RSW1	-	Thuthuka

My packet tracer topology



Question 1

Question A1

Ref	Requirement	Mechanism	Implemented	Verification
R1	ICU medical devices must not initiate connections to the corporate network or internet, only clinical monitoring servers should be able to reach them	Use dedicated VLAN 60 with an extended ACL that permits only clinical monitoring server traffic and blocks all other internal and internet access.	Apply at Sandton MLS1 and MLS2 SVIs	We can ping from an ICU device to the monitoring server, but pings to a corporate Pc and internet test address fail
R2	Provide Wi-Fi for patients and visitors, but it must never reach internal clinical, administrative, or device networks.	Guest VLAN 70 with an ACL blocking internal subnets, plus NAT/PAT and DHCP for guest clients.	Core switches, R1 and WAP1 and WAP2	The guest client does receives an IP address and accesses the internet but cannot reach clinical host.
R3	Clinical workstations must still reach PACS and electronic health record servers during a single core device or link failure, with gateway downtime limited to a few seconds.	Use redundant MLS1/MLS2 pairs with HSRP for gateway failover, EtherChannel for link aggregation, and Rapid PVST+.	Sandton core switches and access switches.	Continuously ping the Electronic health record and PACS servers while disconnecting one core uplink, verify connectivity recovers and the gateway remains available.
R4	Radiology image transfers must not slow clinical workstations or telemedicine consultations during peak times.	Use QoS to prioritize clinical and telemedicine traffic, and rate-limit or shape bulk image transfers where supported.	Core switches and WAN routers	Check quality of service policies and interface counters, then simulate competing traffic to confirm the priority policy works.
R5	Use VLSM to conserve address space and enable WAN summarisation, allowing for two new day clinics within three years.	VLSM with hierarchical site addressing and reserved subnets for two future clinics.	IP addressing plan and routers.	Check subnet calculations and routing tables to confirm future clinic subnets fit within the reserved address space.
R6	All three sites must use dynamic routing, manual route maintenance is not allowed.	Use OSPF with proper network advertisements and route summarisation.	On R1, R2 and R3	Use command such as show ip ospf neighbor and show IP route to verify.

R7	Rustenburg's broadband is unreliable, so connectivity must recover automatically after a link loss no manual intervention.	OSPF must select paths and reconverge correctly over the R2–R3 backup link.	On R1, R2 and R3	Disconnect the R1–R3 link and confirm Rustenburg traffic reroutes through the Polokwane backup path.
R8	All internal hosts must share one public IP for internet access, with only the network edge holding a public address.	Configure NAT overload on R1 and keep private Ips on internal interfaces.	At Sandton R1	You can use show ip nat translations command to verify that multiple internal hosts are able to share the same the same public IP
R9	Device management must be encrypted and centrally authenticated, no plaintext protocols. All devices must log to a central facility and use one authoritative time source.	Use SSHv2 with central AAA, syslog, and NTP, disable Telnet and all other plaintext management services.	On routers, switches and server farm	Check that SSH, AAA authentication, syslog, and time sync work, and confirm Telnet is disabled.
R10	Access layer ports must be protected against unauthorised device connection, rogue DHCP servers and address spoofing. Clinical areas are physically accessible to patients and visitors.	Configure port security, DHCP snooping, Dynamic ARP Inspection, and BPDU Guard, with trusted uplinks set correctly.	On ASW1 to ASW4, PSW1 to PSW2 and RSW1	To verify, connect an unauthorized device or simulate a rogue DHCP server to confirm the relevant port or DHCP protection triggers.
R11	The design must clearly demonstrate compliance with the Group's POPIA responsible-party obligations, especially the section 19 security safeguards.	Implement network segmentation, least privilege ACLs, secure management, central logging, controlled access, and tested backups.	On all sites, core edge and server farm	Verify ACLs, management settings, logs, and backup evidence and test that unauthorized access is blocked.
R12	The full solution must be documented well enough for a third-party engineer to rebuild it and validated with proof it works end to end.	Keep labelled topology, addressing and VLAN docs, saved configs and backups, plus recorded test evidence.	To all devices and technical report	Reopen the saved .pkt file, check the running configurations, and rerun the documented connectivity and failover tests.

A

Question A2

Sandton

VLAN	Network address	Prefix	Usable host range	Broadcast address	Gateway
30 clinicals	10.10.0.0	/24	10.10.0.1–10.10.0.254	10.10.0.255	10.10.0.1
20 Admin	10.10.1.0	/25	10.10.1.1–10.10.1.126	10.10.1.127	10.10.1.1
50 pacs	10.10.1.128	/26	10.10.1.129–10.10.1.190	10.10.1.191	10.10.1.129
60 ICU devices	10.10.1.192	/26	10.10.1.193–10.10.1.254	10.10.1.255	10.10.1.193
90 servers	10.10.2.0	/26	10.10.2.1–10.10.2.62	10.10.2.63	10.10.2.1
10 management	10.10.2.64	/27	10.10.2.65–10.10.2.94	10.10.2.95	10.10.2.65
70 Patient Wi-Fi	10.10.4.0	/23	10.10.4.1–10.10.5.254	10.10.5.255	10.10.4.1
80 – Staff Wi-Fi	10.10.6.0	/23	10.10.6.1–10.10.7.254	10.10.7.255	10.10.6.1

Polokwane

Department	**Network address**	**Prefix**	**Usable host range**	**Broadcast address**	**Gateway**
clinical	10.10.16.0	/25	10.10.16.1-10.10.16.126	10.10.16.127	10.10.16.1
admin	10.10.16.128	/26	10.10.16.129–10.10.16.190	10.10.16.191	10.10.16.129
Medical Devices	10.10.16.192	/27	10.10.16.193–10.10.16.222	10.10.16.223	10.10.16.193
staff Wi-Fi	10.10.16.224	/27	10.10.16.225–10.10.16.254	10.10.16.255	10.10.16.225
Patient Wi-Fi	10.10.17.0	/26	10.10.17.1–10.10.17.62	10.10.17.63	10.10.17.1
Management	10.10.17.64	/27	10.10.17.65–10.10.17.94	10.10.17.95	10.10.17.65

Rustenburg

Department	Network	Prefix	Usable host range	Broadcast address	Gateway
Clinical	10.10.20.0	/26	10.10.20.1–10.10.20.62	10.10.20.63	10.10.20.1
Management	10.10.20.64	27/	10.10.20.65–10.10.20.94	10.10.20.95	10.10.20.65
Patient Wi-Fi	10.10.20.96	27/	10.10.20.97–10.10.20.126	10.10.20.127	10.10.20.97

Staff Wi-Fi	10.10.20.128	27/	10.10.20.129–10.10.20.158	10.10.20.159	10.10.20.129
Medical Devices	10.10.20.160	28/	10.10.20.161–10.10.20.174	10.10.20.175	10.10.20.161

Sandton Router and Internet Edge

Link	Network / Prefix	Usable host range	Broadcast	Gateway / Next hop
R1–MLS1	10.255.0.12/30	10.255.0.13–10.255.0.14	10.255.0.15	MLS1: 10.255.0.13
R1–MLS2	10.255.0.16/30	10.255.0.17–10.255.0.18	10.255.0.19	MLS2: 10.255.0.17
R1–ISP	203.0.113.0/30	203.0.113.1–203.0.113.2	203.0.113.3	ISP: 203.0.113.1

Question A3

I selected Hot standby router protocol (HSRP) for sandton core switches because the network uses cisco multilayer switches and requires a highly available default gateway for clinical workstations accessing the HER and PACS servers.

The alternative considered was VRRP. VRRP is a standard based option that is useful when equipment from different vendors needs to interoperate. However, HSRP is a suitable choice for this cisco based design because it provides an active gateway and a standby gateway that can take over if the active switch fails. This supports the requirement that gateway service should recover within a few seconds. Cisco's high availability design guidance discusses both protocols and identifies HSRP as suitable choice in Cisco environments.

Another alternative was GLBP, which can distribute gateway traffic across multiple routers instead of relying on one active gateway. Although this can improve load sharing, HSRP is simpler to implement and demonstrate for this assignment because the main requirements is gateway redundancy rather than distributing traffic across several gateways. Therefore, HSRP is the preferred option because it meets the availability requirement while keeping the design manageable and both MLS2 and MLS1 must support HSRP. (GeeksforGeeks, 2022) (Networklessons.com, 2026)

Question 2

1. **Basic Switch and Management Configuration from MLS1 to R3**

MLS1 configuration

```
Switch>en
Switch#conf t
Enter configuration commands, one per line.  End with CNTL/Z.
Switch(config)#hostname MLS1
MLS1(config)#ip domain-name thuthuka.local
MLS1(config)#no ip domain-lookup
MLS1(config)#ip routing
MLS1(config)#enable secret Thuthuka
MLS1(config)#service password-encryption
MLS1(config)#username admin privilege 15 secret Admin@2026
MLS1(config)#banner motd #Unauthorised access is not allowed.#
MLS1(config)#line console 0
MLS1(config-line)#password Thukhuka
MLS1(config-line)#login
MLS1(config-line)#logging synchronous
MLS1(config-line)#exec-timeout 5 0
MLS1(config-line)#exit
MLS1(config)#line vty 0 15
MLS1(config-line)#login local
MLS1(config-line)#transport input ssh
MLS1(config-line)#exec-timeout 5 0
MLS1(config-line)#exit
MLS1(config)#ip ssh version 2
Please create RSA keys (of at least 768 bits size) to enable SSH v2.
MLS1(config)#crypto key generate rsa 1024
                                     ^
% Invalid input detected at '^' marker.

MLS1(config)#crypto key generate rsa
The name for the keys will be: MLS1.thuthuka.local
Choose the size of the key modulus in the range of 360 to 2048 for your
  General Purpose Keys. Choosing a key modulus greater than 512 may take
  a few minutes.

How many bits in the modulus [512]: 1024
% Generating 1024 bit RSA keys, keys will be non-exportable...[OK]

MLS1(config)#end
*Mar 19 8:32:31.191: %SSH-5-ENABLED: SSH 2 has been enabled
MLS1#
```

MLS2 configuration

```
Switch#conf t
Enter configuration commands, one per line.  End with CNTL/Z.
Switch(config)#hostname MLS2
MLS2(config)#ip domain-name thuthuka.local
MLS2(config)#no ip domain-lookup
MLS2(config)#ip routing
MLS2(config)#enable secret Thuthuka
MLS2(config)#service password-encryption
MLS2(config)#username admin privilege 15 secret Admin@2026
MLS2(config)#banner motd# Unauthorised access is not allowed.#
                         ^
% Invalid input detected at '^' marker.

MLS2(config)#banner motd #Unauthorised access is not allowed.#
MLS2(config)#line console 0
MLS2(config-line)#password Thuthuka
MLS2(config-line)#login
MLS2(config-line)#logging synchronous
MLS2(config-line)#exec-timeout 5 0
MLS2(config-line)#exit
MLS2(config)#line vty 0 15
MLS2(config-line)#login local
MLS2(config-line)#transport input ssh
MLS2(config-line)#exec-timeout 5 0
MLS2(config-line)#exit
MLS2(config)#ip ssh version 2
Please create RSA keys (of at least 768 bits size) to enable SSH v2.
MLS2(config)#crypto key generate rsa 1024
                                     ^
% Invalid input detected at '^' marker.

MLS2(config)#crypto key generate rsa
The name for the keys will be: MLS2.thuthuka.local
Choose the size of the key modulus in the range of 360 to 2048 for your
  General Purpose Keys. Choosing a key modulus greater than 512 may take
  a few minutes.

How many bits in the modulus [512]: 1024
% Generating 1024 bit RSA keys, keys will be non-exportable...[OK]
```

ASW1

```
Switch>en
Switch#conf t
Enter configuration commands, one per line.  End with CNTL/Z.
Switch(config)#hostname ASW1
ASW1(config)#ip domain-name thuthuka.local
ASW1(config)#no ip domain-lookup
ASW1(config)#enable secret Thuthuka
ASW1(config)#service password-encryption
ASW1(config)#username admin privilege 15 secret Admin@2026
ASW1(config)#banner motd # Unauthorised access is not allowed.#
ASW1(config)#line console 0
ASW1(config-line)#password Thuthuka
ASW1(config-line)#login
ASW1(config-line)#logging synchronous
ASW1(config-line)#exec-timeout 5 0
ASW1(config-line)#exit
ASW1(config)#line vty 0 15
ASW1(config-line)#login local
ASW1(config-line)#transport input ssh
ASW1(config-line)#exec-timeout 5 0
ASW1(config-line)#exit
ASW1(config)#ip ssh version 2
Please create RSA keys (of at least 768 bits size) to enable SSH v2.
ASW1(config)#crypto key generate rsa1024
                                       ^
% Invalid input detected at '^' marker.

ASW1(config)#crypto key generate rsa
The name for the keys will be: ASW1.thuthuka.local
Choose the size of the key modulus in the range of 360 to 4096 for your
  General Purpose Keys. Choosing a key modulus greater than 512 may take
  a few minutes.

How many bits in the modulus [512]: 1024
% Generating 1024 bit RSA keys, keys will be non-exportable...[OK]

ASW1(config)#end
*Mar 19 8:40:32.186: %SSH-5-ENABLED: SSH 2 has been enabled
```

ASW2

```
Switch>enable
Switch#configure terminal
Enter configuration commands, one per line.  End with CNTL/Z.
Switch(config)#hostname ASW2
ASW2(config)#ip domain-name thuthuka.local
ASW2(config)#no ip domain-lookup
ASW2(config)#enable secret Thuthuka
ASW2(config)#service password-encryption
ASW2(config)#username admin privilege 15 secret Admin@2026
ASW2(config)#banner motd #Unauthorised access is not allowed#
ASW2(config)#line console 0
ASW2(config-line)#password Thuthuka
ASW2(config-line)#login
ASW2(config-line)#logging synchronous
ASW2(config-line)#exec-timeout 5 0
ASW2(config-line)#exit
ASW2(config)#line vty 0 15
ASW2(config-line)#login local
ASW2(config-line)#transport input ssh
ASW2(config-line)#exec-timeout 5 0
ASW2(config-line)#exit
ASW2(config)#ip ssh version 2
Please create RSA keys (of at least 768 bits size) to enable SSH v2.
ASW2(config)#crypto key generate rsa
The name for the keys will be: ASW2.thuthuka.local
Choose the size of the key modulus in the range of 360 to 4096 for your
  General Purpose Keys. Choosing a key modulus greater than 512 may take
  a few minutes.

How many bits in the modulus [512]: 1024
% Generating 1024 bit RSA keys, keys will be non-exportable...[OK]

ASW2(config)#end
*Mar 19 8:44:22.298: %SSH-5-ENABLED: SSH 2 has been enabled
ASW2#
%SYS-5-CONFIG_I: Configured from console by console

ASW2#wr
Building configuration...
```

ASW3

```
Switch>enable
Switch#configure terminal
Enter configuration commands, one per line.  End with CNTL/Z.
Switch(config)#hostname ASW3
ASW3(config)#ip domain-name thuthuka.local
ASW3(config)#no ip domain-lookup
ASW3(config)#enable secret Thuthuka
ASW3(config)#service password-encryption
ASW3(config)#username admin privilege 15 secret Admin@2026
ASW3(config)#banner motd #Unauthorised access is not allowed#
ASW3(config)#line console 0
ASW3(config-line)#password Thuthuka
ASW3(config-line)#login
ASW3(config-line)#logging synchronous
ASW3(config-line)#exec-timeout 5 0
ASW3(config-line)#exit
ASW3(config)#line vty 0 15
ASW3(config-line)#login local
ASW3(config-line)#transport input ssh
ASW3(config-line)#exec-timeout 5 0
ASW3(config-line)#exit
ASW3(config)#ip ssh version 2
Please create RSA keys (of at least 768 bits size) to enable SSH v2.
ASW3(config)#crypto key generate rsa
The name for the keys will be: ASW3.thuthuka.local
Choose the size of the key modulus in the range of 360 to 4096 for your
  General Purpose Keys. Choosing a key modulus greater than 512 may take
  a few minutes.

How many bits in the modulus [512]: 1024
% Generating 1024 bit RSA keys, keys will be non-exportable...[OK]

ASW3(config)#end
*Mar 19 8:47:11.985: %SSH-5-ENABLED: SSH 2 has been enabled
ASW3#
%SYS-5-CONFIG_I: Configured from console by console

ASW3#wr
Building configuration...
```

ASW4

```
Switch>enable
Switch#configure terminal
Enter configuration commands, one per line.  End with CNTL/Z.
Switch(config)#hostname ASW4
ASW4(config)#ip domain-name thuthuka.local
ASW4(config)#no ip domain-lookup
ASW4(config)#enable secret Thuthuka
ASW4(config)#service password-encryption
ASW4(config)#username admin privilege 15 secret Admin@2026
ASW4(config)#banner #Unauthorised access is not allowed#
                    ^
% Invalid input detected at '^' marker.

ASW4(config)#banner motd #Unauthorised access is not allowed#
ASW4(config)#line console 0
ASW4(config-line)#password Thuthuka
ASW4(config-line)#login
ASW4(config-line)#logging synchronous
ASW4(config-line)#exec-timeout 5 0
ASW4(config-line)#exit
ASW4(config)#line vty 0 15
ASW4(config-line)#login local
ASW4(config-line)#transport input ssh
ASW4(config-line)#exec-timeout 5 0
ASW4(config-line)#exit
ASW4(config)#ip ssh version 2
Please create RSA keys (of at least 768 bits size) to enable SSH v2.
ASW4(config)#crypto key generate rsa
The name for the keys will be: ASW4.thuthuka.local
Choose the size of the key modulus in the range of 360 to 4096 for your
  General Purpose Keys. Choosing a key modulus greater than 512 may take
  a few minutes.

How many bits in the modulus [512]: 1024
% Generating 1024 bit RSA keys, keys will be non-exportable...[OK]

ASW4(config)#end
*Mar 19 8:50:43.146: %SSH-5-ENABLED: SSH 2 has been enabled
ASW4#
%SYS-5-CONFIG_I: Configured from console by console

ASW4#wr
Building configuration...
[OK]
```

R1

```
Router>enable
Router#configure terminal
Enter configuration commands, one per line.  End with CNTL/Z.
Router(config)#hostname Sandton R1
                               ^
% Invalid input detected at '^' marker.

Router(config)#hostname R1
R1(config)#ip domain-name thuthuka.local
R1(config)#no ip domain-lookup
R1(config)#enable secret Thuthuka
R1(config)#service password-encryption
R1(config)#username admin privilege 15 secret Admin@2026
R1(config)#banner motd #Unauthorised access is not allowed#
R1(config)#line console 0
R1(config-line)#password Thuthuka
R1(config-line)#login
R1(config-line)#logging synchronous
R1(config-line)#exec-timeout 5 0
R1(config-line)#exit
R1(config)#line vty 0 15
R1(config-line)#login local
R1(config-line)#transport input ssh
R1(config-line)#exec-timeout 5 0
R1(config-line)#exit
R1(config)#ip ssh version 2
Please create RSA keys (of at least 768 bits size) to enable SSH v2.
R1(config)#crypto key generate rsa
The name for the keys will be: R1.thuthuka.local
Choose the size of the key modulus in the range of 360 to 4096 for your
  General Purpose Keys. Choosing a key modulus greater than 512 may take
  a few minutes.

How many bits in the modulus [512]: 1024
% Generating 1024 bit RSA keys, keys will be non-exportable...[OK]

R1(config)#end
*Mar 19 8:8:4.739: %SSH-5-ENABLED: SSH 2 has been enabled
R1#
%SYS-5-CONFIG_I: Configured from console by console
```

R2

```
Router>enable
Router#configure terminal
Enter configuration commands, one per line.  End with CNTL/Z.
Router(config)#hostname R2
R2(config)#ip domain-name thuthuka.local
R2(config)#no ip domain-lookup
R2(config)#enable secret Thuthuka
R2(config)#service password-encryption
R2(config)#username admin privilege 15 secret Admin@2026
R2(config)#banner motd #Unauthorised access is prohibited.#
R2(config)#line console 0
R2(config-line)#password Thuthuka
R2(config-line)#login
R2(config-line)#logging synchronous
R2(config-line)#exec-timeout 5 0
R2(config-line)#exit
R2(config)#line vty 0 15
R2(config-line)#login local
R2(config-line)#transport input ssh
R2(config-line)#exec-timeout 5 0
R2(config-line)#exit
R2(config)#ip ssh version 2
Please create RSA keys (of at least 768 bits size) to enable SSH v2.
R2(config)#crypto key generate rsa
The name for the keys will be: R2.thuthuka.local
Choose the size of the key modulus in the range of 360 to 4096 for your
  General Purpose Keys. Choosing a key modulus greater than 512 may take
  a few minutes.

How many bits in the modulus [512]: 1024
% Generating 1024 bit RSA keys, keys will be non-exportable...[OK]

R2(config)#end
*Mar 19 8:13:33.936: %SSH-5-ENABLED: SSH 2 has been enabled
R2#
%SYS-5-CONFIG_I: Configured from console by console

R2#
R2#wr
Building configuration
```

R3

```
Router>enable
Router#configure terminal
Enter configuration commands, one per line.  End with CNTL/Z.
Router(config)#hostname R3
R3(config)#ip domain-name thuthuka.local
R3(config)#no ip domain-lookup
R3(config)#enable secret Thuthuka
R3(config)#service password-encryption
R3(config)#username admin privilege 15 secret Admin@2026
R3(config)#banner motd #Unauthorised access is not allowed#
R3(config)#line console 0
R3(config-line)#password Thuthuka
R3(config-line)#login
R3(config-line)#logging synchronous
R3(config-line)#exec-timeout 5 0
R3(config-line)#exit
R3(config)#line vty 0 15
R3(config-line)#login local
R3(config-line)#transport input ssh
R3(config-line)#exec-timeout 5 0
R3(config-line)#exit
R3(config)#ip ssh version 2
Please create RSA keys (of at least 768 bits size) to enable SSH v2.
R3(config)#crypto key generate rsa
The name for the keys will be: R3.thuthuka.local
Choose the size of the key modulus in the range of 360 to 4096 for your
  General Purpose Keys. Choosing a key modulus greater than 512 may take
  a few minutes.

How many bits in the modulus [512]: 1024
% Generating 1024 bit RSA keys, keys will be non-exportable...[OK]

R3(config)#end
*Mar 19 8:17:26.538: %SSH-5-ENABLED: SSH 2 has been enabled
R3#
%SYS-5-CONFIG_I: Configured from console by console

R3#
R3#wr
Building configuration
```

2. Configure VLANs and VTP

MLS1 VLANs configuration

```
MLS1>en
Password:
MLS1#conf t
Enter configuration commands, one per line.  End with CNTL/Z.
MLS1(config)#vlan 10
MLS1(config-vlan)#name MANAGEMENT
MLS1(config-vlan)#exit
MLS1(config)#vlan 20
MLS1(config-vlan)#name ADMIN
MLS1(config-vlan)#exit
MLS1(config)#vlan 30
MLS1(config-vlan)#name CLINICAL
MLS1(config-vlan)#exit
MLS1(config)#vlan 40
MLS1(config-vlan)#name MEDICAL-DEVICES
MLS1(config-vlan)#exit
MLS1(config)#vlan 50
MLS1(config-vlan)#name PACS
MLS1(config-vlan)#exit
MLS1(config)#vlan 60
MLS1(config-vlan)#name ICU-DEVICES
MLS1(config-vlan)#exit
MLS1(config)#vlan 70
MLS1(config-vlan)#name GUEST-WIFI
MLS1(config-vlan)#exit
MLS1(config)#vlan 80
MLS1(config-vlan)#name STAFF-WIFI
MLS1(config-vlan)#exit
MLS1(config)#vlan 90
MLS1(config-vlan)#name SERVERS
MLS1(config-vlan)#exit
MLS1(config)#vlan 100
MLS1(config-vlan)#name VOICE
MLS1(config-vlan)#exit
MLS1(config)#vlan 999
MLS1(config-vlan)#name PARKING-NATIVE
MLS1(config-vlan)#exit
MLS1(config)#end
MLS1#
```

Configuring MLS1 as the VTP server

```
MLS1#vtp domain THUTHUKA
         ^
% Invalid input detected at '^' marker.

MLS1#vtp mode server
         ^
% Invalid input detected at '^' marker.

MLS1#vtp version 2
         ^
% Invalid input detected at '^' marker.

MLS1#
MLS1#
MLS1#
MLS1#
MLS1#
MLS1#conf t
Enter configuration commands, one per line.  End with CNTL/Z.
MLS1(config)#vtp domain THUTHUKA
Changing VTP domain name from NULL to THUTHUKA
MLS1(config)#vtp mode server
Device mode already VTP SERVER.
MLS1(config)#vtp version 2%SW_VLAN-6-VTP_DOMAIN_NAME_CHG: VTP domain name changed to THUTHUKA.

MLS1(config)#vtp version 2
MLS1(config)#vtp password Thuthuka
Setting device VLAN database password to Thuthuka
MLS1(config)#end
MLS1#
%SYS-5-CONFIG_I: Configured from console by console

MLS1#copy running-config startup-config
Destination filename [startup-config]?
Building configuration...
[OK]
MLS1#wr
Building configuration...
[OK]
MLS1#
```

Configuring MLS2 as the VTP client

```
60   ICU-DEVICES                      active
70   GUEST-WIFI                       active
80   STAFF-WIFI                       active
90   SERVERS                          active
100  VOICE                            active
999  PARKING-NATIVE                   active
1002 fddi-default                     active
1003 token-ring-default               active
1004 fddinet-default                  active
1005 trnet-default                    active
MLS2#
MLS2#
MLS2#
MLS2#
MLS2#
MLS2#
MLS2#
MLS2#
MLS2#
MLS2#
MLS2#
MLS2#show vtp status
VTP Version capable             : 1 to 2
VTP version running             : 2
VTP Domain Name                 : THUTHUKA
VTP Pruning Mode                : Disabled
VTP Traps Generation            : Disabled
Device ID                       : 00E0.A322.CE00
Configuration last modified by 0.0.0.0 at 3-19-93 09:29:16

Feature VLAN :
--------------
VTP Operating Mode                : Client
Maximum VLANs supported locally   : 1005
Number of existing VLANs          : 16
Configuration Revision            : 1
MD5 digest                        : 0x1D 0xA4 0x70 0x4C 0x90 0x33 0x48 0xFE
                                    0xB8 0xCB 0x09 0xED 0x73 0x3F 0x5B 0x6B
MLS2#
```

One advantage and one risk of using VTP server/client mode in this environment.

Advantage of VTP is that it makes vlans management easir because vlans only need to be created on MLS1, which acts as the server. MLS2 will then receives the Vlan information automatically, saving time and keeping vlans configurations consistent across both switches. (NetworkLessons.com, 2013)

However, VTP also has a risk, if an incorrect VLAN configuration is made on the server, it can be distributed to client switches and affect network connectivity. This could cause several departments to lose access to their network services. Therefore, changes to vlan configurations should be checked carefully and regular configuration backups should be kept. (NetworkLessons.com, 2013)

3. Configure Spanning Tree and Layer 2 Protection

Mls1 configuration Spanning Tree

```
User Access Verification

Password:

MLS1>en
Password:
MLS1#enable
MLS1#configure terminal
Enter configuration commands, one per line.  End with CNTL/Z.
MLS1(config)#line console 0
MLS1(config-line)#password Thuthuka
MLS1(config-line)#login
MLS1(config-line)#end
MLS1#copy running-config startup-config
%SYS-5-CONFIG_I: Configured from console by console

MLS1#copy running-config startup-config
Destination filename [startup-config]?
Building configuration...
[OK]
MLS1#
MLS1#
MLS1#
MLS1#enable
MLS1#configure terminal
Enter configuration commands, one per line.  End with CNTL/Z.
MLS1(config)#spanning-tree mode rapid-pvst
MLS1(config)#end
MLS1#
%SYS-5-CONFIG_I: Configured from console by console

MLS1#configure terminal
Enter configuration commands, one per line.  End with CNTL/Z.
MLS1(config)#spanning-tree vlan 10,20,30,50 root primary
MLS1(config)#spanning-tree vlan 60,70,80,90 root secondary
MLS1(config)#end
MLS1#
%SYS-5-CONFIG_I: Configured from console by console

MLS1#
```

Configure Spanning Tree and Layer 2 Protection on MLS2

```
Unauthorised access is not allowed.

User Access Verification

Password:

MLS2>en
Password:
MLS2#configure terminal
Enter configuration commands, one per line.  End with CNTL/Z.
MLS2(config)#spanning-tree mode rapid-pvst
MLS2(config)#end
MLS2#
%SYS-5-CONFIG_I: Configured from console by console

MLS2#enable
MLS2#configure terminal
Enter configuration commands, one per line.  End with CNTL/Z.
MLS2(config)#spanning-tree vlan 10,20,30,50 root secondary
MLS2(config)#spanning-tree vlan 60,70,80,90 root primary
MLS2(config)#end
MLS2#
%SYS-5-CONFIG_I: Configured from console by console

MLS2#wr
Building configuration...
[OK]
MLS2#
```

Configure Spanning Tree and Layer 2 Protection on ASW1

```
ASW1#
ASW1#
ASW1#
ASW1#
ASW1#en
ASW1#conf t
Enter configuration commands, one per line.  End with CNTL/Z.
ASW1(config)#spanning-tree mode rapid-pvst
ASW1(config)#
ASW1(config)#interface fastEthernet 0/1
ASW1(config-if)#no spanning-tree portfast
ASW1(config-if)#no spanning-tree bpduguard enable
ASW1(config-if)#spanning-tree guard root
ASW1(config-if)#exit
ASW1(config)#
ASW1(config)#interface range fastEthernet 0/2-24
ASW1(config-if-range)#spanning-tree portfast
```

```
%Portfast has been configured on FastEthernet0/24 but will only
have effect when the interface is in a non-trunking mode.
ASW1(config-if-range)#spanning-tree bpduguard enable
ASW1(config-if-range)#exit
ASW1(config)#
ASW1(config)#end
ASW1#copy running-config startup-config
%SYS-5-CONFIG_I: Configured from console by console
```

Configure Spanning Tree and Layer 2 Protection on ASW2

```
ASW2>en
Password:
ASW2#conf t
Enter configuration commands, one per line.  End with CNTL/Z.
ASW2(config)#spanning-tree mode rapid-pvst
ASW2(config)#
ASW2(config)#interface fastEthernet 0/1
ASW2(config-if)#no spanning-tree portfast
ASW2(config-if)#no spanning-tree bpduguard enable
ASW2(config-if)#spanning-tree guard root
ASW2(config-if)#exit
ASW2(config)#
ASW2(config)#interface range fastEthernet 0/2-24
ASW2(config-if-range)#spanning-tree portfast
```

```
%Portfast has been configured on FastEthernet0/24 but will only
have effect when the interface is in a non-trunking mode.
ASW2(config-if-range)#spanning-tree bpduguard enable
ASW2(config-if-range)#exit
ASW2(config)#
ASW2(config)#end
ASW2#copy running-config startup-config
%SYS-5-CONFIG_I: Configured from console by console
```

Configure Spanning Tree and Layer 2 Protection on ASW3

```
ASW3>en
Password:
ASW3#enable
ASW3#show spanning-tree summary
Switch is in rapid-pvst mode
Root bridge for: STAFF-WIFI
Extended system ID           is enabled
Portfast Default             is disabled
PortFast BPDU Guard Default  is disabled
Portfast BPDU Filter Default is disabled
Loopguard Default            is disabled
EtherChannel misconfig guard is disabled
UplinkFast                   is disabled
BackboneFast                 is disabled
Configured Pathcost method used is short

Name                   Blocking Listening Learning Forwarding STP Active
---------------------- -------- --------- -------- ---------- ----------
VLAN0010                      3         0        0          1          4
VLAN0030                      3         0        0          1          4
VLAN0070                      0         0        0          4          4
VLAN0080                      3         0        0          1          4
VLAN0999                      3         0        0          1          4

---------------------- -------- --------- -------- ---------- ----------
12 vlans                     12         0        0          8         20

ASW3#

!
spanning-tree mode rapid-pvst
spanning-tree extend system-id
!
interface FastEthernet0/1
!
interface FastEthernet0/2
 spanning-tree portfast
 spanning-tree bpduguard enable
!
interface FastEthernet0/3
 spanning-tree portfast
 spanning-tree bpduguard enable
!
interface FastEthernet0/4
 spanning-tree portfast
 spanning-tree bpduguard enable
!
```

Configure Spanning Tree and Layer 2 Protection on ASW3

```
ASW4>en
Password:
ASW4#conf t
Enter configuration commands, one per line.  End with CNTL/Z.
ASW4(config)#spanning-tree mode rapid-pvst
ASW4(config)#
ASW4(config)#interface fastEthernet 0/1
ASW4(config-if)#no spanning-tree portfast
ASW4(config-if)#no spanning-tree bpduguard enable
ASW4(config-if)#no spanning-tree guard root
ASW4(config-if)#no shutdown
ASW4(config-if)#exit
ASW4(config)#
ASW4(config)#interface range fastEthernet 0/2-24
ASW4(config-if-range)#spanning-tree portfast
%Portfast has been configured on FastEthernet0/24 but will only
have effect when the interface is in a non-trunking mode.
ASW4(config-if-range)#spanning-tree bpduguard enable
ASW4(config-if-range)#exit
ASW4(config)#
ASW4(config)#end
ASW4#copy running-config startup-config
%SYS-5-CONFIG_I: Configured from console by console
```

4. Configure EtherChannel

Configured a layer 2 LACP EtherChannel and configured port channel as an 802.1Q trunk on MLS1

```
MLS1#show etherchannel summary
Flags:  D - down        P - in port-channel
        I - stand-alone s - suspended
        H - Hot-standby (LACP only)
        R - Layer3      S - Layer2
        U - in use      f - failed to allocate aggregator
        u - unsuitable for bundling
        w - waiting to be aggregated
        d - default port


Number of channel-groups in use: 1
Number of aggregators:           1

Group  Port-channel  Protocol    Ports
------+-------------+-----------+-----------------------------------------------

1      Po1(SU)           LACP   Fa0/4(P) Fa0/5(P)
MLS1#show interfaces trunk
Port        Mode         Encapsulation  Status        Native vlan
Po1         on           802.1q         trunking      999
Fa0/3       on           802.1q         trunking      999

Port        Vlans allowed on trunk
Po1         10,20,30,50,60,70,80,90,999
Fa0/3       10,20,30,50,60,70,80,90,999

Port        Vlans allowed and active in management domain
Po1         10,20,30,50,60,70,80,90,999
Fa0/3       10,20,30,50,60,70,80,90,999

Port        Vlans in spanning tree forwarding state and not pruned
Po1         10,20,30,50,60,70,80,90,999
Fa0/3       10,20,30,50,60,70,80,90,999
```

Configured a layer 2 LACP EtherChannel and configured port channel as an 802.1Q trunk on MLS2

```
MLS2>en
Password:
MLS2#show etherchannel summary
Flags:  D - down        P - in port-channel
        I - stand-alone s - suspended
        H - Hot-standby (LACP only)
        R - Layer3      S - Layer2
        U - in use      f - failed to allocate aggregator
        u - unsuitable for bundling
        w - waiting to be aggregated
        d - default port


Number of channel-groups in use: 1
Number of aggregators:           1

Group  Port-channel  Protocol    Ports
------+-------------+-----------+-----------------------------------------------

1      Po1(SU)           LACP   Fa0/3(P) Fa0/4(P)
MLS2#show interfaces trunk
Port        Mode         Encapsulation  Status        Native vlan
Po1         on           802.1q         trunking      999

Port        Vlans allowed on trunk
Po1         10,20,30,50,60,70,80,90,999

Port        Vlans allowed and active in management domain
Po1         10,20,30,50,60,70,80,90,999

Port        Vlans in spanning tree forwarding state and not pruned
Po1         10,20,30,50,60,70,80,90,999

MLS2#
```

This method distributes network traffic across the EtherChannel links based on the source and destination IP addresses. It is suitable for the hospital because multiple devices communicate with clinical systems, PACS servers and other hospital servers. This helps distribute traffic across the available links, improve bandwidth utilisation and maintain connectivity if one physical link fails. (Ivan, 2022)

5.Configure Trunk Links

Configure Trunk Links on MLS1

```
MLS1>en
Password:
MLS1#show interfaces trunk
Port        Mode         Encapsulation  Status        Native vlan
Po1         on           802.1q         trunking      999
Fa0/1       on           802.1q         trunking      999
Fa0/2       on           802.1q         trunking      999
Fa0/3       on           802.1q         trunking      999

Port        Vlans allowed on trunk
Po1         10,20,30,50,60,70,80,90,999
Fa0/1       10,30,70,80,999
Fa0/2       10,30,999
Fa0/3       10,20,30,50,60,70,80,90,999

Port        Vlans allowed and active in management domain
Po1         10,20,30,50,60,70,80,90,999
Fa0/1       10,30,70,80,999
Fa0/2       10,30,999
Fa0/3       10,20,30,50,60,70,80,90,999

Port        Vlans in spanning tree forwarding state and not pruned
Po1         10,20,30,50,60,70,80,90,999
Fa0/1       10,30,70,80,999
Fa0/2       10,30,999
Fa0/3       10,20,30,50,60,70,80,90,999
```

Configure Trunk Links on MLS2

```
Password:

MLS2>en
Password:
MLS2#show interfaces trunk
Port        Mode         Encapsulation  Status        Native vlan
Po1         on           802.1q         trunking      999
Fa0/1       on           802.1q         trunking      999
Fa0/2       on           802.1q         trunking      999

Port        Vlans allowed on trunk
Po1         10,20,30,50,60,70,80,90,999
Fa0/1       10,30,999
Fa0/2       10,30,70,80,999

Port        Vlans allowed and active in management domain
Po1         10,20,30,50,60,70,80,90,999
Fa0/1       10,30,999
Fa0/2       10,30,70,80,999

Port        Vlans in spanning tree forwarding state and not pruned
Po1         10,20,30,50,60,70,80,90,999
Fa0/1       10,30,999
Fa0/2       10,30,70,80,999
```

Disable Dynamic Trunking Protocol negotiation and Configure 802.1Q trunks on ASW1

```
Password:

ASW1>en
Password:
ASW1#configure terminal
Enter configuration commands, one per line.  End with CNTL/Z.
ASW1(config)#interface range fa0/2-24
ASW1(config-if-range)#switchport mode access
ASW1(config-if-range)#switchport nonegotiate
ASW1(config-if-range)#end
ASW1#copy running-config startup-config
%SYS-5-CONFIG_I: Configured from console by console

ASW1#copy running-config startup-config
Destination filename [startup-config]?
Building configuration...
[OK]
ASW1#
ASW1#
```

```
ASW1>en
Password:
ASW1#show interfaces trunk
Port        Mode         Encapsulation  Status        Native vlan
Fa0/1       on           802.1q         trunking      999

Port        Vlans allowed on trunk
Fa0/1       10,30,70,80,999

Port        Vlans allowed and active in management domain
Fa0/1       10,30,70,80,999

Port        Vlans in spanning tree forwarding state and not pruned
Fa0/1       10,30,70,80,999

ASW1#
```

Disable Dynamic Trunking Protocol negotiation and Configure 802.1Q trunks on ASW2

```
Password:

ASW2>en
Password:
ASW2#conf t
Enter configuration commands, one per line.  End with CNTL/Z.
ASW2(config)#interface range fa0/2-24
ASW2(config-if-range)#switchport mode access
ASW2(config-if-range)#switchport nonegotiate
ASW2(config-if-range)#end
ASW2#copy running-config startup-config
%SYS-5-CONFIG_I: Configured from console by console

ASW2#copy running-config startup-config
Destination filename [startup-config]?
Building configuration...
[OK]
ASW2#
ASW2#
```

```
ASW2>en
Password:
ASW2#show interfaces trunk
Port        Mode         Encapsulation  Status        Native vlan
Fa0/1       on           802.1q         trunking      999

Port        Vlans allowed on trunk
Fa0/1       10,30,999

Port        Vlans allowed and active in management domain
Fa0/1       10,30,999

Port        Vlans in spanning tree forwarding state and not pruned
Fa0/1       10,30,999
```

Disable Dynamic Trunking Protocol negotiation and Configure 802.1Q trunks on ASW3

```
Password:

ASW3>en
Password:
ASW3#conf t
Enter configuration commands, one per line.  End with CNTL/Z.
ASW3(config)#interface range fa0/2-24
ASW3(config-if-range)#switchport mode access
ASW3(config-if-range)#switchport nonegotiate
ASW3(config-if-range)#end
ASW3#copy running-config startup-config
%SYS-5-CONFIG_I: Configured from console by console

ASW3#copy running-config startup-config
Destination filename [startup-config]?
Building configuration...
[OK]
ASW3#
```

```
ASW3>en
Password:
ASW3#enable
ASW3#show interfaces trunk
Port        Mode         Encapsulation  Status        Native vlan
Fa0/1       on           802.1q         trunking      999

Port        Vlans allowed on trunk
Fa0/1       10,30,70,80,999

Port        Vlans allowed and active in management domain
Fa0/1       10,30,70,80,999

Port        Vlans in spanning tree forwarding state and not pruned
Fa0/1       10,30,70,80,999
```

Disable Dynamic Trunking Protocol negotiation and Configure 802.1Q trunks on ASW4

```
ASW4#show interfaces switchport
Name: Fa0/1
Switchport: Enabled
Administrative Mode: trunk
Operational Mode: trunk
Administrative Trunking Encapsulation: dot1q
Operational Trunking Encapsulation: dot1q
Negotiation of Trunking: Off
Access Mode VLAN: 1 (default)
Trunking Native Mode VLAN: 999 (PARKING-NATIVE)
```

```
ASW4>en
Password:
ASW4#enable
ASW4#show interfaces trunk
Port        Mode         Encapsulation  Status        Native vlan
Fa0/1       on           802.1q         trunking      999

Port        Vlans allowed on trunk
Fa0/1       10,30,70,80,999

Port        Vlans allowed and active in management domain
Fa0/1       10,30,70,80,999

Port        Vlans in spanning tree forwarding state and not pruned
Fa0/1       10,30,70,80,999
```

6. Configure Inter-VLAN Routing and HSRP

Configure Inter-VLAN Routing and HSRP on MLS1

```
MLS1#
MLS1#configure terminal
Enter configuration commands, one per line.  End with CNTL/Z.
MLS1(config)#ip routing
MLS1(config)#interface gigabitEthernet0/1
MLS1(config-if)#no switchport
MLS1(config-if)#ip address 10.255.0.13 255.255.255.252
MLS1(config-if)#no shutdown
MLS1(config-if)#end
MLS1#copy running-config startup-config
%SYS-5-CONFIG_I: Configured from console by console

MLS1#copy running-config startup-config
Destination filename [startup-config]?
Building configuration...
[OK]
```

. Configure Inter-VLAN Routing and HSRP on MLS2

```
MLS2#
MLS2#configure terminal
Enter configuration commands, one per line.  End with CNTL/Z.
MLS2(config)#ip routing
MLS2(config)#interface gigabitEthernet0/1
MLS2(config-if)#no switchport
MLS2(config-if)#ip address 10.255.0.17 255.255.255.252
MLS2(config-if)#no shutdown
MLS2(config-if)#end
MLS2#copy running-config startup-config
%SYS-5-CONFIG_I: Configured from console by console

MLS2#copy running-config startup-config
Destination filename [startup-config]?
Building configuration...
[OK]
MLS2#
```

. Configure Inter-VLAN Routing and HSRP R1

```
R1>en
Password:
R1#conf t
Enter configuration commands, one per line.  End with CNTL/Z.
R1(config)#interface gigabitEthernet0/0
R1(config-if)#no switchport
                ^
% Invalid input detected at '^' marker.

R1(config-if)#ip address 10.255.0.14 255.255.255.252
R1(config-if)#no shutdown

R1(config-if)#exit
R1(config)#
R1(config)#interface gigabitEthernet0/1
R1(config-if)#no switchport
                ^
% Invalid input detected at '^' marker.

R1(config-if)#ip address 10.255.0.18 255.255.255.252
R1(config-if)#no shutdown

R1(config-if)#exit
R1(config)#
R1(config)#end
%LINK-5-CHANGED: Interface GigabitEthernet0/0, changed state to up
```

Configured four SVIs on MLS1 and assigned it as the active HSRP switch for VLANs 10, 20 , 30 and 50

```
MLS1>en
Password:
MLS1#conf t
Enter configuration commands, one per line.  End with CNTL/Z.
MLS1(config)#interface vlan 10
MLS1(config-if)#ip address 10.10.2.66 255.255.255.192
MLS1(config-if)#standby version 2
MLS1(config-if)#standby 10 ip 10.10.2.65
MLS1(config-if)#standby 10 priority 110
MLS1(config-if)#standby 10 preempt
MLS1(config-if)#standby 10 track gigabitEthernet0/1 20
                                                     ^
% Invalid input detected at '^' marker.

MLS1(config-if)#no shutdown
MLS1(config-if)#exit
%LINK-5-CHANGED: Interface Vlan10, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface Vlan10, changed state to up

%HSRP-6-STATECHANGE: Vlan10 Grp 10 state Init -> Init

MLS1(config-if)#exit
MLS1(config)#interface vlan 30
MLS1(config-if)#ip address 10.10.0.2 255.255.255.0
MLS1(config-if)#standby version 2
MLS1(config-if)#standby 30 ip 10.10.0.1
MLS1(config-if)#standby 30 priority 110
MLS1(config-if)#standby 30 preempt
MLS1(config-if)#standby 30 track gigabitEthernet0/1 20
                                                     ^
% Invalid input detected at '^' marker.

MLS1(config-if)#no shutdown
MLS1(config-if)#exit
%LINK-5-CHANGED: Interface Vlan30, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface Vlan30, changed state to up
```

```
%LINEPROTO-5-UPDOWN: Line protocol on Interface Vlan30, changed state to up

%HSRP-6-STATECHANGE: Vlan30 Grp 30 state Init -> Init

MLS1(config-if)#exit
MLS1(config)#
%HSRP-6-STATECHANGE: Vlan10 Grp 10 state Speak -> Standby

%HSRP-6-STATECHANGE: Vlan10 Grp 10 state Standby -> Active

MLS1(config)#interface vlan 50
MLS1(config-if)#ip address 10.10.1.130 255.255.255.192
MLS1(config-if)#standby version 2
MLS1(config-if)#standby 50 ip 10.10.1.129
MLS1(config-if)#standby 50 priority 110
MLS1(config-if)#standby 50 preempt
MLS1(config-if)#standby 50 track gigabitEthernet0/1 20
                                                     ^
% Invalid input detected at '^' marker.

MLS1(config-if)#no shutdown
MLS1(config-if)#exit
MLS1(config)#
MLS1(config)#end
%LINK-5-CHANGED: Interface Vlan50, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface Vlan50, changed state to up

%HSRP-6-STATECHANGE: Vlan50 Grp 50 state Init -> Init

MLS1(config)#end
%HSRP-6-STATECHANGE: Vlan30 Grp 30 state Speak -> Standby

%HSRP-6-STATECHANGE: Vlan30 Grp 30 state Standby -> Active

MLS1(config)#end
MLS1#
%SYS-5-CONFIG_I: Configured from console by console
```

Configured Vlan 20 on MLS1

```
MLS1>en
Password:
MLS1#conf t
Enter configuration commands, one per line.  End with CNTL/Z.
MLS1(config)#interface vlan 20
MLS1(config-if)#ip address 10.10.1.2 255.255.255.128
MLS1(config-if)#standby version 2
MLS1(config-if)#standby 20 ip 10.10.1.1
MLS1(config-if)#standby 20 priority 110
MLS1(config-if)#standby 20 preempt
MLS1(config-if)#no shutdown
MLS1(config-if)#exit
MLS1(config)#
MLS1(config)#end
MLS1#
%SYS-5-CONFIG_I: Configured from console by console

MLS1#show standby brief
                     P indicates configured to preempt.
                     |
Interface   Grp  Pri P State    Active          Standby         Virtual IP
Vl10        10   110 P Active   local           unknown         10.10.2.65
Vl30        30   110 P Active   local           unknown         10.10.0.1
Vl50        50   110 P Active   local           unknown         10.10.1.129
Vl20        20   110 P Speak    10.10.1.2       unknown         10.10.1.1
MLS1#
```

Configured the VLAN 60 SVI on MLS2 and assigned MLS2 as the active HSRP switch for the ICU Devices VLAN, using virtual gateway 10.10.1.193.

```
MLS2>en
Password:
MLS2#conf t
Enter configuration commands, one per line.  End with CNTL/Z.
MLS2(config)#interface vlan 60
MLS2(config-if)#ip address 10.10.1.194 255.255.255.192
MLS2(config-if)#standby version 2
MLS2(config-if)#standby 60 ip 10.10.1.193
MLS2(config-if)#standby 60 priority 110
MLS2(config-if)#standby 60 preempt
MLS2(config-if)#no shutdown
MLS2(config-if)#exit
MLS2(config)#
%LINK-5-CHANGED: Interface Vlan60, changed state to up
```

Configured the VLAN 70 SVI on MLS2 and assigned MLS2 as the active HSRP switch for the Guest Wi-Fi VLAN, using virtual gateway 10.10.4.1.

```
MLS2#
MLS2#configure terminal
Enter configuration commands, one per line.  End with CNTL/Z.
MLS2(config)#
MLS2(config)#interface vlan 70
MLS2(config-if)#ip address 10.10.4.3 255.255.254.0
MLS2(config-if)#standby version 2
MLS2(config-if)#standby 70 ip 10.10.4.1
MLS2(config-if)#standby 70 priority 110
MLS2(config-if)#standby 70 preempt
MLS2(config-if)#no shutdown
MLS2(config-if)#exit
MLS2(config)#
MLS2(config)#end
%LINK-5-CHANGED: Interface Vlan70, changed state to up
```

Configured the VLAN 80 SVI on MLS2 and assigned MLS2 as the active HSRP switch for the Staff Wi-Fi VLAN, using virtual gateway 10.10.6.1.

```
MLS2#
MLS2#configure terminal
Enter configuration commands, one per line.  End with CNTL/Z.
MLS2(config)#
MLS2(config)#interface vlan 80
MLS2(config-if)#ip address 10.10.6.3 255.255.255.0
MLS2(config-if)#standby version 2
MLS2(config-if)#standby 80 ip 10.10.6.1
MLS2(config-if)#standby 80 priority 110
MLS2(config-if)#standby 80 preempt
MLS2(config-if)#no shutdown
MLS2(config-if)#exit
MLS2(config)#
MLS2(config)#end
%LINK-5-CHANGED: Interface Vlan80, changed state to up
```

Configured the VLAN 90 SVI on MLS2 and assigned MLS2 as the active HSRP switch for the Servers VLAN, using virtual gateway IP address 10.10.2.1 and priority 110

```
MLS2#configure terminal
Enter configuration commands, one per line.  End with CNTL/Z.
MLS2(config)#
MLS2(config)#interface vlan 90
MLS2(config-if)#ip address 10.10.2.3 255.255.255.192
MLS2(config-if)#standby version 2
MLS2(config-if)#standby 90 ip 10.10.2.1
MLS2(config-if)#standby 90 priority 110
MLS2(config-if)#standby 90 preempt
MLS2(config-if)#no shutdown
MLS2(config-if)#exit
MLS2(config)#
MLS2(config)#end
%LINK-5-CHANGED: Interface Vlan90, changed state to up
```

Configured the VLAN 10, 20, 30 and 50 SVIs on MLS2 and set up HSRP with priority 100, allowing MLS2 to act as the standby switch while MLS1 remains the preferred active switch for these VLANs

```
MLS2#
MLS2#configure terminal
Enter configuration commands, one per line.  End with CNTL/Z.
MLS2(config)#
MLS2(config)#interface vlan 10
MLS2(config-if)#ip address 10.10.2.67 255.255.255.192
MLS2(config-if)#standby version 2
MLS2(config-if)#standby 10 ip 10.10.2.65
MLS2(config-if)#standby 10 priority 100
MLS2(config-if)#standby 10 preempt
MLS2(config-if)#no shutdown
MLS2(config-if)#exit
MLS2(config)#
MLS2(config)#interface vlan 20
MLS2(config-if)#ip address 10.10.1.3 255.255.255.128
MLS2(config-if)#standby version 2
MLS2(config-if)#standby 20 ip 10.10.1.1
MLS2(config-if)#standby 20 priority 100
MLS2(config-if)#standby 20 preempt
MLS2(config-if)#no shutdown
MLS2(config-if)#exit
MLS2(config)#
MLS2(config)#interface vlan 30
MLS2(config-if)#ip address 10.10.0.3 255.255.255.0
MLS2(config-if)#standby version 2
MLS2(config-if)#standby 30 ip 10.10.0.1
MLS2(config-if)#standby 30 priority 100
MLS2(config-if)#standby 30 preempt
MLS2(config-if)#no shutdown
MLS2(config-if)#exit
MLS2(config)#
MLS2(config)#interface vlan 50
MLS2(config-if)#ip address 10.10.1.131 255.255.255.192
MLS2(config-if)#standby version 2
MLS2(config-if)#standby 50 ip 10.10.1.129
MLS2(config-if)#standby 50 priority 100
MLS2(config-if)#standby 50 preempt
MLS2(config-if)#no shutdown
MLS2(config-if)#exit
MLS2(config)#
```

```
MLS2#
MLS2#show standby brief
                     P indicates configured to preempt.
                     |
Interface   Grp  Pri P State    Active          Standby         Virtual IP
V160        60   110 P Active   local           unknown         10.10.1.193
V170        70   110 P Active   local           unknown         10.10.4.1
V180        80   110 P Active   local           unknown         10.10.6.1
V190        90   110 P Active   local           unknown         10.10.2.1
V110        10   100 P Standby  10.10.2.66      local           10.10.2.65
V120        20   100 P Standby  10.10.1.2       local           10.10.1.1
V130        30   100 P Standby  10.10.0.2       local           10.10.0.1
V150        50   100 P Standby  10.10.1.130     local           10.10.1.129
MLS2#
```

Configured the VLAN 60, 70, 80 and 90 SVIs on MLS1 and established standby HSRP groups with priority 100, allowing MLS2 to remain the preferred active switch for these VLANs.

```
MLS1#configure terminal
Enter configuration commands, one per line.  End with CNTL/Z.
MLS1(config)#
MLS1(config)#interface vlan 60
MLS1(config-if)#ip address 10.10.1.195 255.255.255.192
MLS1(config-if)#standby version 2
MLS1(config-if)#standby 60 ip 10.10.1.193
MLS1(config-if)#standby 60 priority 100
MLS1(config-if)#standby 60 preempt
MLS1(config-if)#no shutdown
MLS1(config-if)#exit
MLS1(config)#
MLS1(config)#interface vlan 70
MLS1(config-if)#ip address 10.10.4.2 255.255.254.0
MLS1(config-if)#standby version 2
MLS1(config-if)#standby 70 ip 10.10.4.1
MLS1(config-if)#standby 70 priority 100
MLS1(config-if)#standby 70 preempt
MLS1(config-if)#no shutdown
MLS1(config-if)#exit
MLS1(config)#
MLS1(config)#interface vlan 80
MLS1(config-if)#ip address 10.10.6.2 255.255.255.0
MLS1(config-if)#standby version 2
MLS1(config-if)#standby 80 ip 10.10.6.1
MLS1(config-if)#standby 80 priority 100
MLS1(config-if)#standby 80 preempt
MLS1(config-if)#no shutdown
MLS1(config-if)#exit
MLS1(config)#
MLS1(config)#interface vlan 90
MLS1(config-if)#ip address 10.10.2.2 255.255.255.192
MLS1(config-if)#standby version 2
MLS1(config-if)#standby 90 ip 10.10.2.1
MLS1(config-if)#standby 90 priority 100
MLS1(config-if)#standby 90 preempt
MLS1(config-if)#no shutdown
MLS1(config-if)#exit
MLS1(config)#
```

```
MLS1#show standby brief
                     P indicates configured to preempt.
                     |
Interface   Grp  Pri P State    Active          Standby         Virtual IP
Vl10        10   110 P Active   local           10.10.2.67      10.10.2.65
Vl30        30   110 P Active   local           10.10.0.3       10.10.0.1
Vl50        50   110 P Active   local           10.10.1.131     10.10.1.129
Vl20        20   110 P Active   local           10.10.1.3       10.10.1.1
Vl60        60   100 P Standby  10.10.1.194     local           10.10.1.193
Vl70        70   100 P Standby  10.10.4.3       local           10.10.4.1
Vl80        80   100 P Standby  10.10.6.3       local           10.10.6.1
Vl90        90   100 P Standby  10.10.2.3       local           10.10.2.1
MLS1#
```

Configured HSRP interface tracking on MLS1 for VLANs 10, 20, 30, 50, 60, 70, 80 and 90. Each HSRP group monitors the MLS1 routed uplink, GigabitEthernet0/1, connected to R1. This allows HSRP to respond if the uplink goes down and supports gateway failover to MLS2.

```
MLS1#
MLS1#configure terminal
Enter configuration commands, one per line.  End with CNTL/Z.
MLS1(config)#
MLS1(config)#interface vlan 10
MLS1(config-if)#standby 10 track GigabitEthernet0/1
MLS1(config-if)#exit
MLS1(config)#
MLS1(config)#interface vlan 20
MLS1(config-if)#standby 20 track GigabitEthernet0/1
MLS1(config-if)#exit
MLS1(config)#
MLS1(config)#interface vlan 50
MLS1(config-if)#standby 50 track GigabitEthernet0/1
MLS1(config-if)#exit
MLS1(config)#
MLS1(config)#interface vlan 60
MLS1(config-if)#standby 60 track GigabitEthernet0/1
MLS1(config-if)#exit
MLS1(config)#
MLS1(config)#interface vlan 70
MLS1(config-if)#standby 70 track GigabitEthernet0/1
MLS1(config-if)#exit
MLS1(config)#
MLS1(config)#interface vlan 80
MLS1(config-if)#standby 80 track GigabitEthernet0/1
MLS1(config-if)#exit
MLS1(config)#
MLS1(config)#interface vlan 90
MLS1(config-if)#standby 90 track GigabitEthernet0/1
MLS1(config-if)#exit
MLS1(config)#
MLS1(config)#end
MLS1#
%SYS-5-CONFIG_I: Configured from console by console

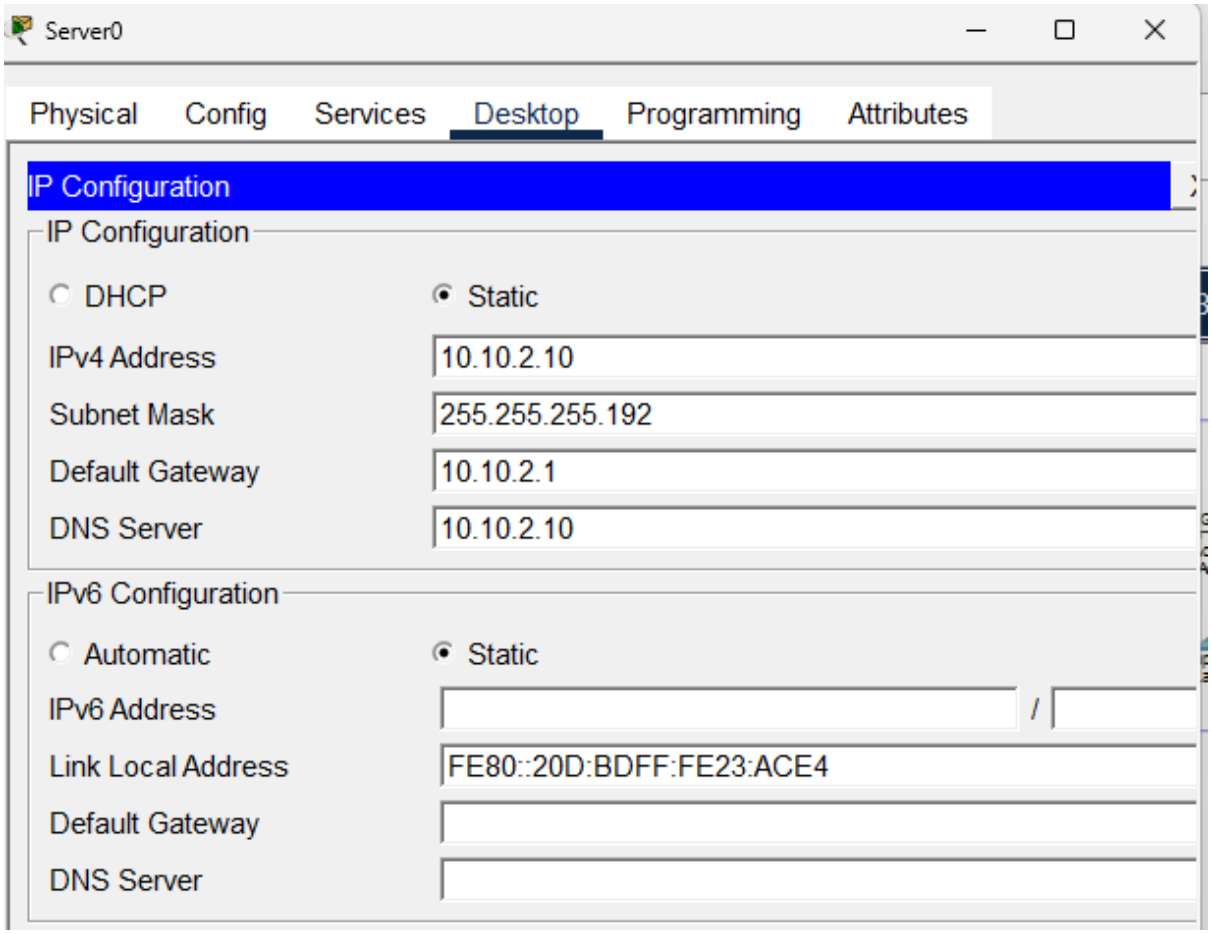
MLS1#wr
Building configuration...
[OK]
```

Configured HSRP interface tracking on MLS2 for VLANs 10 (Management), 20 (Admin), 30 (Clinical), 50 (PACS), 60 (ICU Devices), 70 (Guest Wi-Fi), 80 (Staff Wi-Fi) and 90 (Servers). Each HSRP group monitors MLS2's routed uplink, GigabitEthernet0/1, connected to R1. This allows HSRP to respond if the uplink fails and supports gateway failover to MLS1.

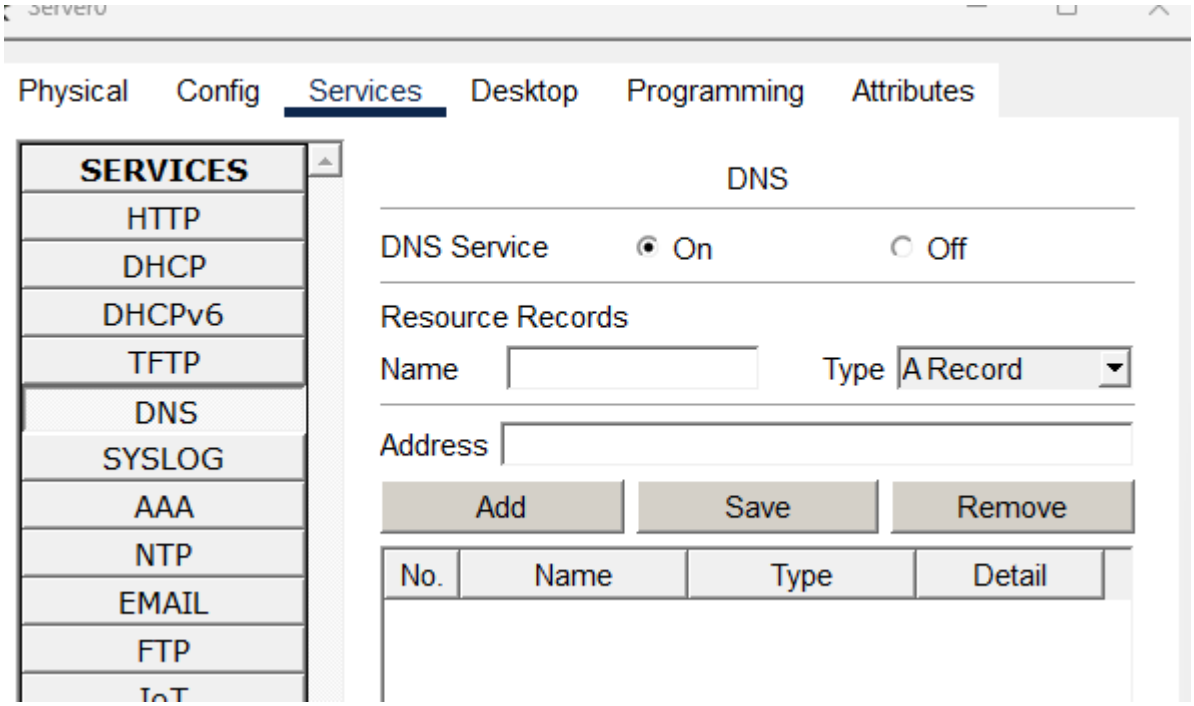
```
MLS2>en
Password:
MLS2#configure terminal
Enter configuration commands, one per line.  End with CNTL/Z.
MLS2(config)#
MLS2(config)#interface vlan 10
MLS2(config-if)#standby 10 track GigabitEthernet0/1
MLS2(config-if)#exit
MLS2(config)#
MLS2(config)#interface vlan 20
MLS2(config-if)#standby 20 track GigabitEthernet0/1
MLS2(config-if)#exit
MLS2(config)#
MLS2(config)#interface vlan 30
MLS2(config-if)#standby 30 track GigabitEthernet0/1
MLS2(config-if)#exit
MLS2(config)#
MLS2(config)#interface vlan 50
MLS2(config-if)#standby 50 track GigabitEthernet0/1
MLS2(config-if)#exit
MLS2(config)#
MLS2(config)#interface vlan 60
MLS2(config-if)#standby 60 track GigabitEthernet0/1
MLS2(config-if)#exit
MLS2(config)#
MLS2(config)#interface vlan 70
MLS2(config-if)#standby 70 track GigabitEthernet0/1
MLS2(config-if)#exit
MLS2(config)#
MLS2(config)#interface vlan 80
MLS2(config-if)#standby 80 track GigabitEthernet0/1
MLS2(config-if)#exit
MLS2(config)#
MLS2(config)#interface vlan 90
MLS2(config-if)#standby 90 track GigabitEthernet0/1
MLS2(config-if)#exit
MLS2(config)#
MLS2(config)#end
MLS2#
```

7. Configure DHCP Services

Server IP Configuration, The Sandton DNS server is assigned a static IP address in VLAN 90. The default gateway is set to the HSRP virtual IP address to provide a reliable gateway, and the server is configured to use itself for DNS services.



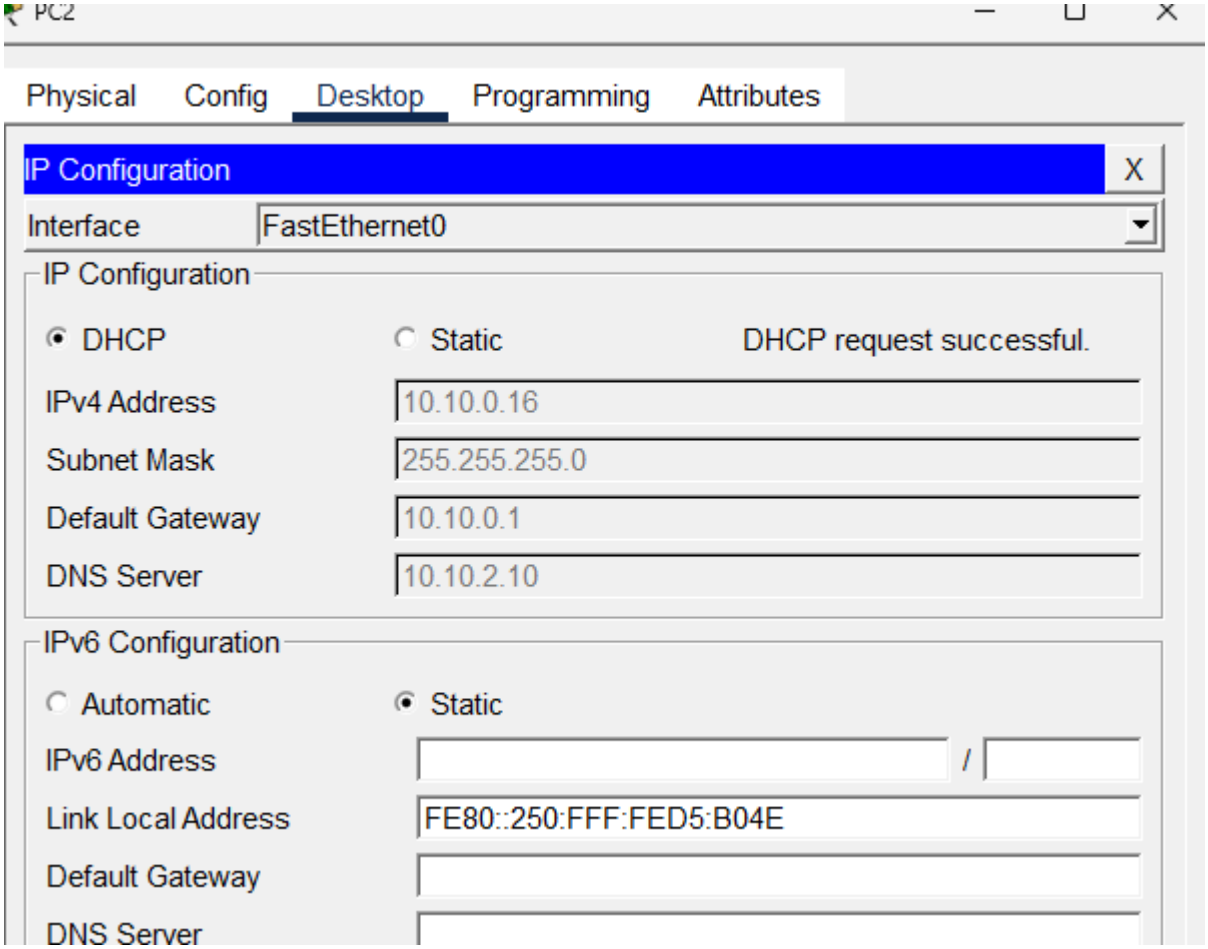
The DNS service is enabled on the Sandton server to provide name resolution for devices on the Thuthuka network.



Clinical DHCP Pool, A DHCP pool is configured for VLAN 30 to automatically assign IP addresses to clinical devices. The first 15 addresses are excluded, and the HSRP virtual IP and internal DNS server are configured on MLS1.

```
MLS1>en
Password:
MLS1#conf t
Enter configuration commands, one per line.  End with CNTL/Z.
MLS1(config)#ip dhcp excluded-address 10.10.0.1 10.10.0.15
MLS1(config)#
MLS1(config)#ip dhcp pool CLINICAL
MLS1(dhcp-config)#network 10.10.0.0 255.255.255.0
MLS1(dhcp-config)#default-router 10.10.0.1
MLS1(dhcp-config)#dns-server 10.10.2.10
MLS1(dhcp-config)#exit
MLS1(config)#
MLS1(config)#end
MLS1#
%SYS-5-CONFIG_I: Configured from console by console

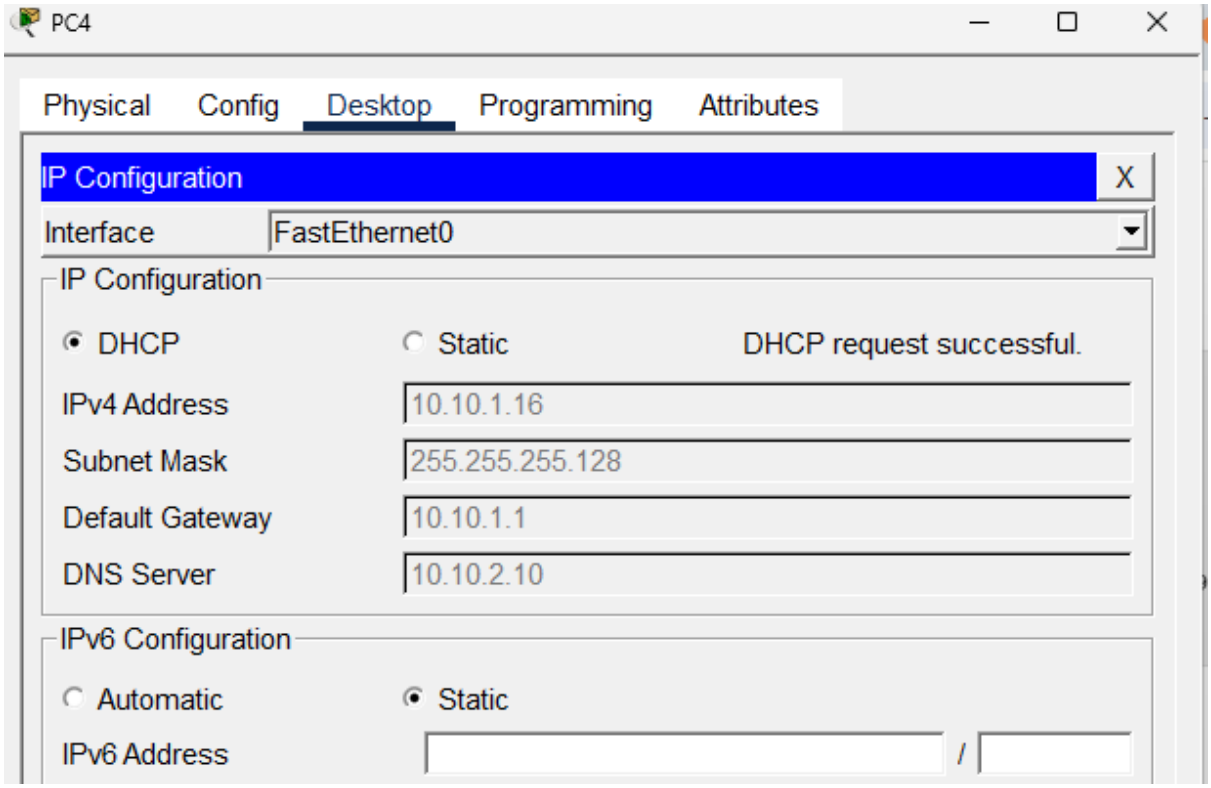
MLS1#wr
Building configuration...
[OK]
MLS1#
```



Administration DHCP Pool – DHCP is configured for VLAN 20 to automatically assign IP addresses to administrative devices. The first 15 addresses are excluded, with the HSRP virtual gateway and internal DNS server provided to clients.

```
[OK]
MLS1#configure terminal
Enter configuration commands, one per line.  End with CNTL/Z.
MLS1(config)#
MLS1(config)#ip dhcp excluded-address 10.10.1.1 10.10.1.15
MLS1(config)#
MLS1(config)#ip dhcp pool ADMIN
MLS1(dhcp-config)#network 10.10.1.0 255.255.255.128
MLS1(dhcp-config)#default-router 10.10.1.1
MLS1(dhcp-config)#dns-server 10.10.2.10
MLS1(dhcp-config)#exit
MLS1(config)#
MLS1(config)#end
MLS1#
%SYS-5-CONFIG_I: Configured from console by console

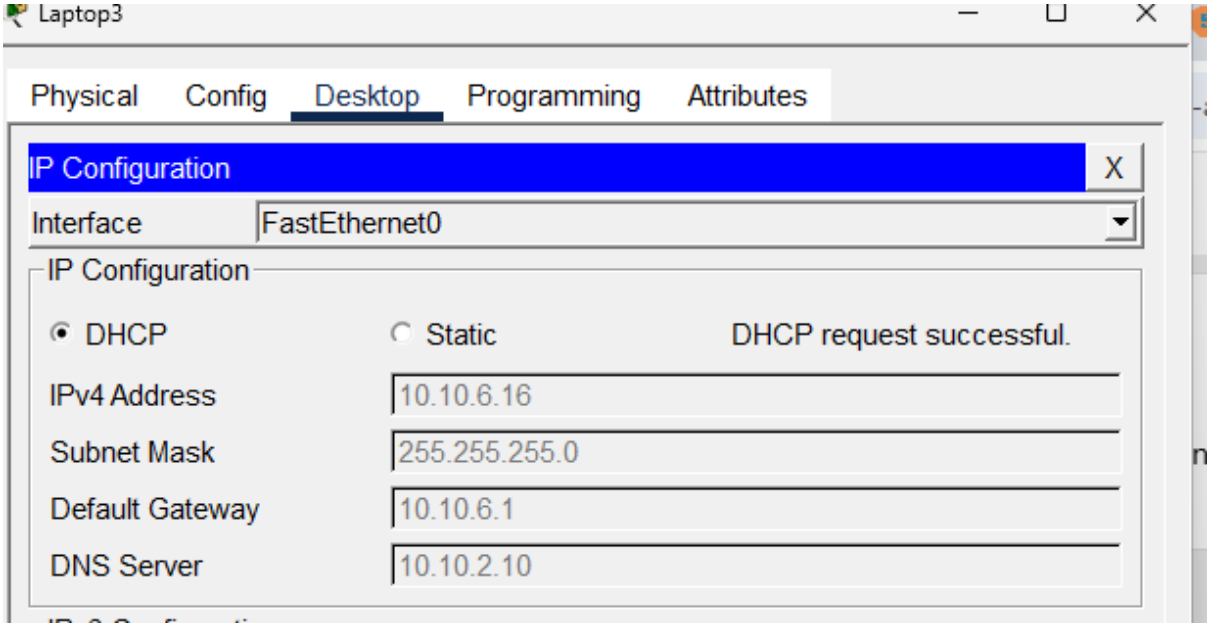
MLS1#wr
Building configuration...
[OK]
```



Staff Wi-Fi DHCP Pool – A DHCP pool is configured for VLAN 80 to automatically assign IP addresses to staff wireless devices. The first 15 addresses are excluded, and the HSRP virtual gateway and internal DNS server are provided

```
MLS1#wr
Building configuration...
[OK]
MLS1#configure terminal
Enter configuration commands, one per line.  End with CNTL/Z.
MLS1(config)#
MLS1(config)#ip dhcp excluded-address 10.10.6.1 10.10.6.15
MLS1(config)#
MLS1(config)#ip dhcp pool STAFF-WIFI
MLS1(dhcp-config)#network 10.10.6.0 255.255.255.0
MLS1(dhcp-config)#default-router 10.10.6.1
MLS1(dhcp-config)#dns-server 10.10.2.10
MLS1(dhcp-config)#exit
MLS1(config)#
MLS1(config)#end
MLS1#
%SYS-5-CONFIG_I: Configured from console by console

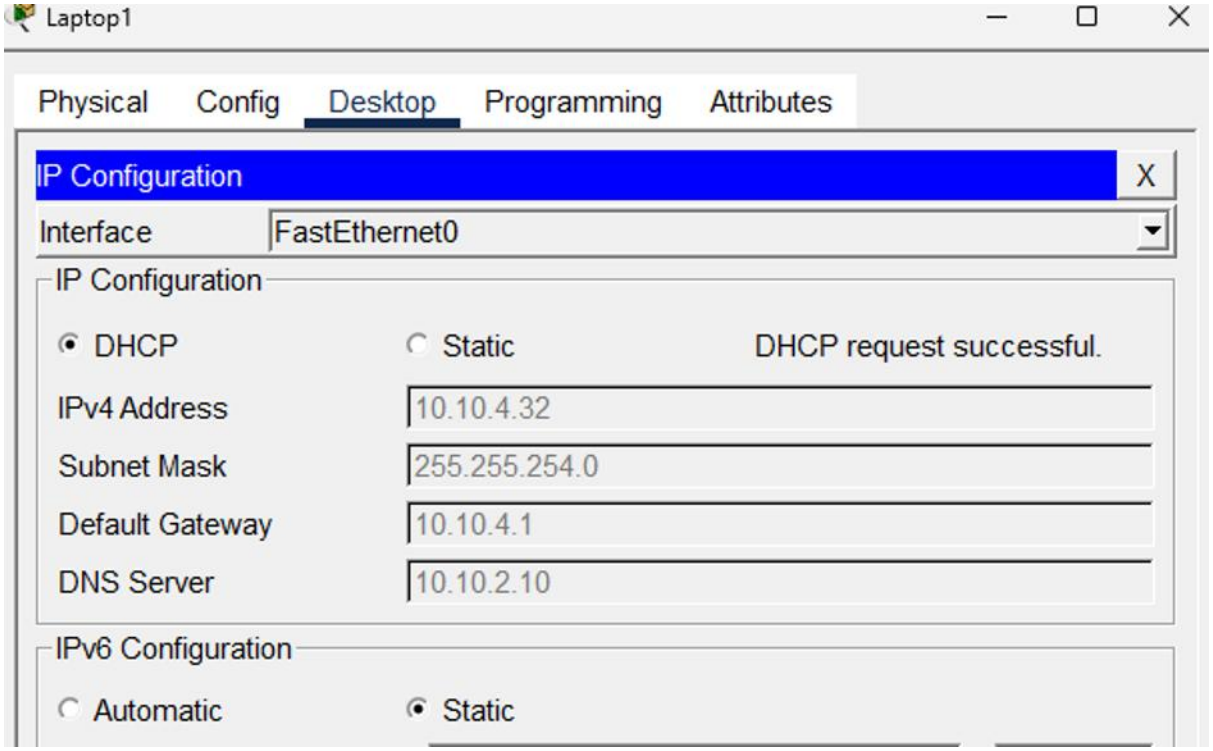
MLS1#wr
Building configuration...
```



Patient/Guest Wi-Fi DHCP Pool – DHCP is configured for VLAN 70 to automatically assign IP addresses to patient and guest wireless devices. The first 15 addresses are excluded, and the HSRP virtual gateway and internal DNS server are provided.

```
MLS1#wr
Building configuration...
[OK]
MLS1#configure terminal
Enter configuration commands, one per line.  End with CNTL/Z.
MLS1(config)#
MLS1(config)#ip dhcp excluded-address 10.10.4.1 10.10.4.15
MLS1(config)#
MLS1(config)#ip dhcp pool GUEST-WIFI
MLS1(dhcp-config)#network 10.10.4.0 255.255.254.0
MLS1(dhcp-config)#default-router 10.10.4.1
MLS1(dhcp-config)#dns-server 10.10.2.10
MLS1(dhcp-config)#exit
MLS1(config)#
MLS1(config)#end
MLS1#
%SYS-5-CONFIG_I: Configured from console by console

MLS1#wr
Building configuration...
[OK]
MLS1#
```



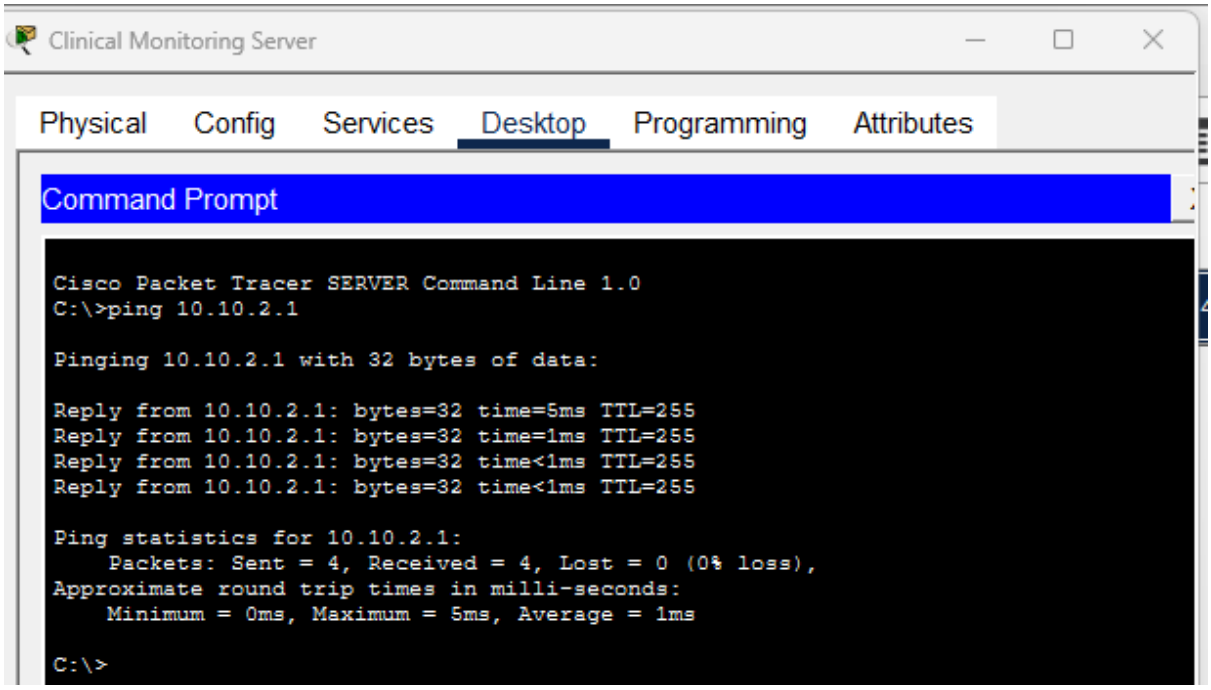
Why Medical Devices Were Excluded from DHCP

The medical device VLAN was excluded from dynamic DHCP because devices such as ICU monitors and infusion pumps require consistent IP address to communicate reliably with clinical monitoring systems. Using static IP addresses from changing unexpectedly. This also makes it easier to apply access control rules that restrict medical devices to authorised monitoring.

Question 3

3.1 Medical Device Isolation

The Clinical Monitoring Server (10.10.2.11) successfully communicates with its HSRP default gateway (10.10.2.1), confirming connectivity to the Server VLAN.



Configured the MEDICAL-DEVICE-ISOLATION ACL to allow VLAN 60 devices to communicate with the Clinical Monitoring Server (10.10.2.11) while blocking access to all other destinations. Applied the ACL inbound on VLAN 60 and saved the configuration.

```
Password:

MLS1>en
Password:
MLS1#conf t
Enter configuration commands, one per line.  End with CNTL/Z.
MLS1(config)#ip access-list extended MEDICAL-DEVICE-ISOLATION
MLS1(config-ext-nacl)# permit ip 10.10.1.192 0.0.0.63 host 10.10.2.11
MLS1(config-ext-nacl)# deny ip 10.10.1.192 0.0.0.63 any
MLS1(config-ext-nacl)#exit
MLS1(config)#
MLS1(config)#interface vlan 60
MLS1(config-if)# ip access-group MEDICAL-DEVICE-ISOLATION in
MLS1(config-if)#exit
MLS1(config)#
MLS1(config)#end
MLS1#write memory
%SYS-5-CONFIG_I: Configured from console by console

MLS1#write memory
Building configuration...
[OK]
```

Configured the same MEDICAL-DEVICE-ISOLATION ACL on MLS2 to maintain the same security policy during HSRP failover. Applied it inbound on VLAN 60 and saved the configuration.

```
MLS2>en
Password:
MLS2#conf t
Enter configuration commands, one per line.  End with CNTL/Z.
MLS2(config)#ip access-list extended MEDICAL-DEVICE-ISOLATION
MLS2(config-ext-nacl)# permit ip 10.10.1.192 0.0.0.63 host 10.10.2.11
MLS2(config-ext-nacl)# deny ip 10.10.1.192 0.0.0.63 any
MLS2(config-ext-nacl)#exit
MLS2(config)#
MLS2(config)#interface vlan 60
MLS2(config-if)# ip access-group MEDICAL-DEVICE-ISOLATION in
MLS2(config-if)#exit
MLS2(config)#
MLS2(config)#end
MLS2#write memory
%SYS-5-CONFIG_I: Configured from console by console

MLS2#write memory
Building configuration...
[OK]
MLS2#
MLS2#
```

The ACL is applied in the inbound on the vlan 60 so that traffic from medical devices is checked as it enters the multilayer switch. This blocks unauthorised traffic before it can reach other networks. This is better than applying it outbound on multiple Vlan interfaces because it provides one central point of control.

3.2 Patient and Visitor Wireless Isolation

The PATIENT-WIFI-ISOLATION ACL was applied inbound on VLAN 70 to allow DNS and external network traffic while blocking patient access to internal hospital networks on MLS2

```
MLS2>en
Password:
MLS2#enable
MLS2#configure terminal
Enter configuration commands, one per line.  End with CNTL/Z.
MLS2(config)#
MLS2(config)#ip access-list extended PATIENT-WIFI-ISOLATION
MLS2(config-ext-nacl)# permit udp any host 10.10.2.10 eq 53
MLS2(config-ext-nacl)# permit tcp any host 10.10.2.10 eq 53
MLS2(config-ext-nacl)# deny ip any 10.0.0.0 0.255.255.255
MLS2(config-ext-nacl)# deny ip any 172.16.0.0 0.15.255.255
MLS2(config-ext-nacl)# deny ip any 192.168.0.0 0.0.255.255
MLS2(config-ext-nacl)# permit ip any any
MLS2(config-ext-nacl)#
MLS2(config-ext-nacl)#interface vlan 70
MLS2(config-if)# ip access-group PATIENT-WIFI-ISOLATION in
MLS2(config-if)#
MLS2(config-if)#end
MLS2#write memory
%SYS-5-CONFIG_I: Configured from console by console

MLS2#write memory
Building configuration...
[OK]
```

The PATIENT-WIFI-ISOLATION ACL was applied inbound on VLAN 70 to allow DNS and external network traffic while blocking patient access to internal hospital networks on MLS1

```
MLS1>en
Password:
MLS1#enable
MLS1#configure terminal
Enter configuration commands, one per line.  End with CNTL/Z.
MLS1(config)#
MLS1(config)#ip access-list extended PATIENT-WIFI-ISOLATION
MLS1(config-ext-nacl)# permit udp any host 10.10.2.10 eq 53
MLS1(config-ext-nacl)# permit tcp any host 10.10.2.10 eq 53
MLS1(config-ext-nacl)# deny ip any 10.0.0.0 0.255.255.255
MLS1(config-ext-nacl)# deny ip any 172.16.0.0 0.15.255.255
MLS1(config-ext-nacl)# deny ip any 192.168.0.0 0.0.255.255
MLS1(config-ext-nacl)# permit ip any any
MLS1(config-ext-nacl)#
MLS1(config-ext-nacl)#interface vlan 70
MLS1(config-if)# ip access-group PATIENT-WIFI-ISOLATION in
MLS1(config-if)#
MLS1(config-if)#end
MLS1#write memory
%SYS-5-CONFIG_I: Configured from console by console

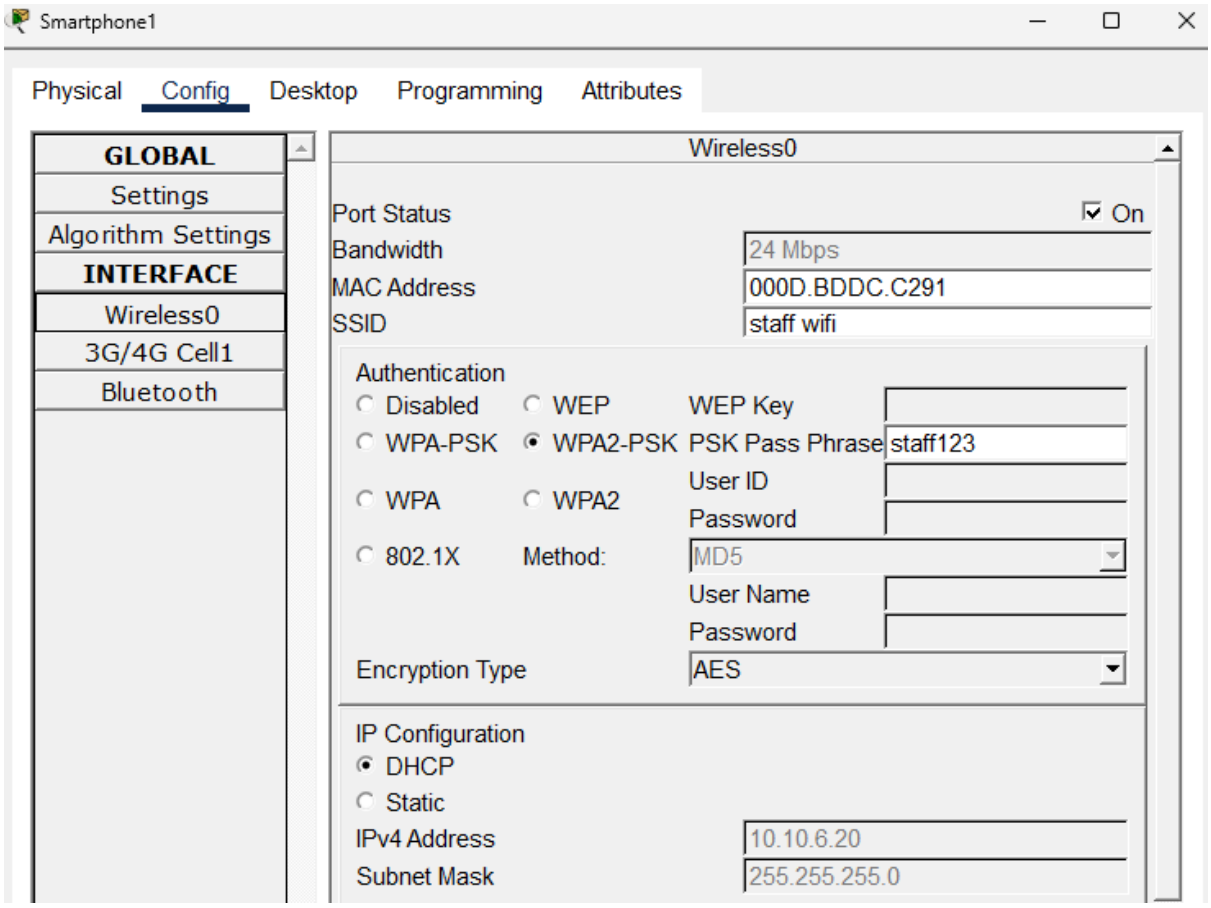
MLS1#write memory
Building configuration...
[OK]
MLS1#
```

The PATIENT-WIFI-ISOLATION ACL was configured on ASW3 and applied inbound on Fa0/4 to allow DNS traffic and restrict patient Wi-Fi access to internal hospital networks.

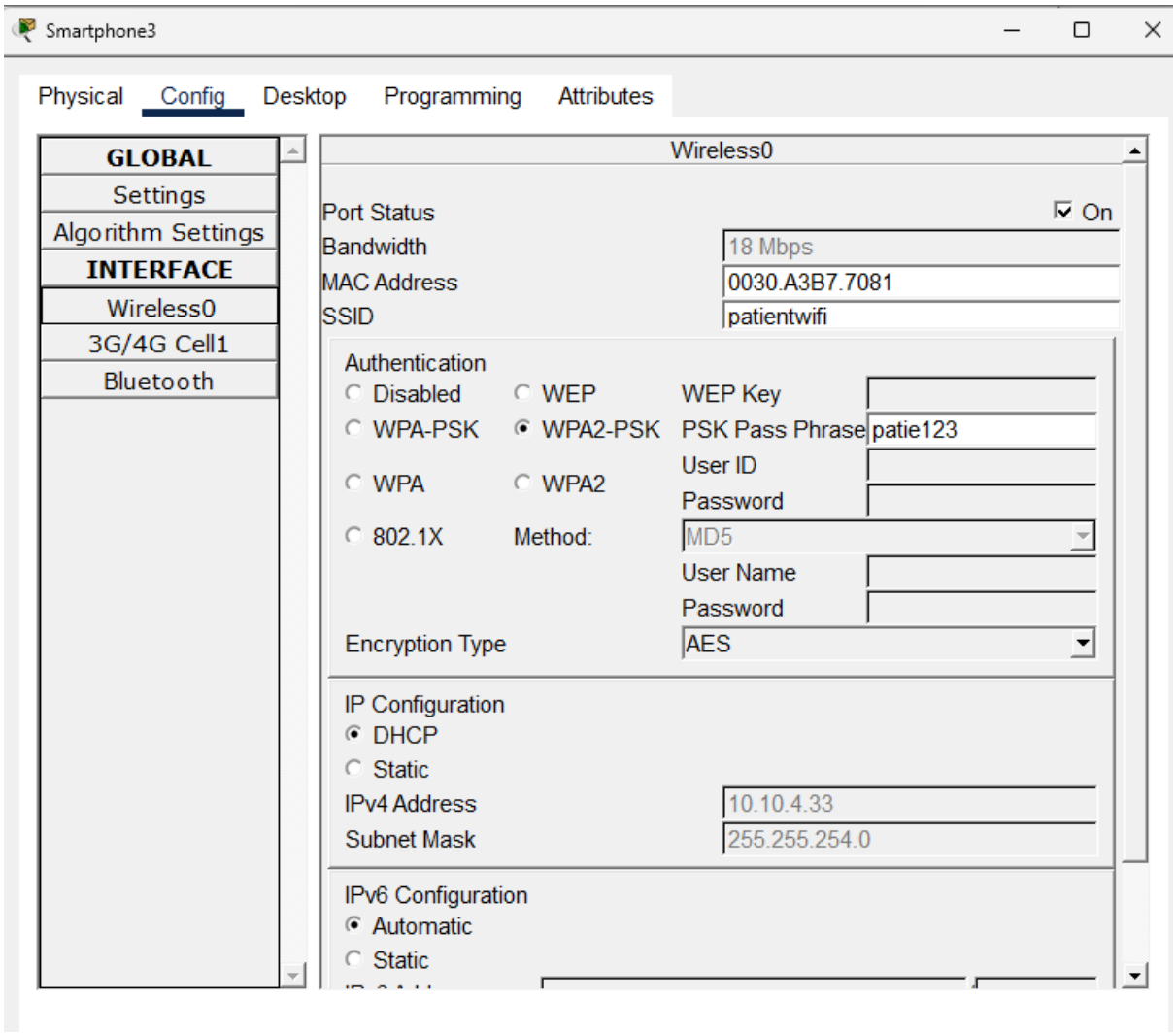
```
ASW3#
ASW3#
ASW3#enable
ASW3#configure terminal
Enter configuration commands, one per line.  End with CNTL/Z.
ASW3(config)#
ASW3(config)#ip access-list extended PATIENT-WIFI-ISOLATION
ASW3(config-ext-nacl)#permit udp any host 10.10.2.10 eq 53
ASW3(config-ext-nacl)#permit tcp any host 10.10.2.10 eq 53
ASW3(config-ext-nacl)#deny ip any 10.0.0.0 0.255.255.255
ASW3(config-ext-nacl)#deny ip any 172.16.0.0 0.15.255.255
ASW3(config-ext-nacl)#deny ip any 192.168.0.0 0.0.255.255
ASW3(config-ext-nacl)#permit ip any any
ASW3(config-ext-nacl)#
ASW3(config-ext-nacl)#end
ASW3#write memory
%SYS-5-CONFIG_I: Configured from console by console

ASW3#write memory
Building configuration...
[OK]
ASW3#
ASW3#
```

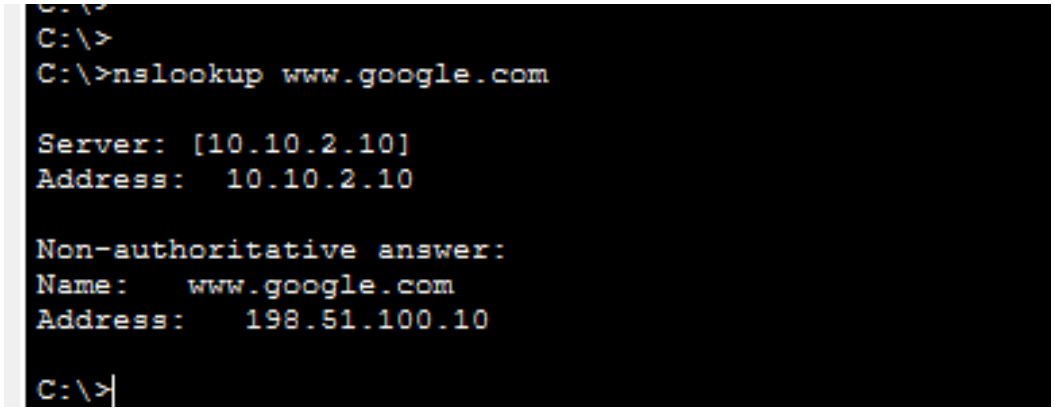
Staff phone connected to staff Wi-Fi with DHCP



Patient phone connected to patient Wi-Fi with DHCP



shows that the patient Wi-Fi device successfully resolved www.google.com to 198.51.100.10 using DNS server 10.10.2.10. This confirms that DNS resolution is working correctly on the patient wireless network.



3.3 Access-Layer Defence in Depth

Configure port security on ASW1

```
ASW1#
ASW1#
ASW1#enable
ASW1#configure terminal
Enter configuration commands, one per line.  End with CNTL/Z.
ASW1(config)#interface range fastEthernet 0/2 - 5
ASW1(config-if-range)#switchport mode access
ASW1(config-if-range)#switchport port-security
ASW1(config-if-range)#switchport port-security maximum 2
ASW1(config-if-range)#switchport port-security mac-address sticky
ASW1(config-if-range)#switchport port-security violation restrict
ASW1(config-if-range)#end
ASW1#
%SYS-5-CONFIG_I: Configured from console by console

ASW1#wr
Building configuration...
[OK]
ASW1#
```

```
ASW1#wr
Building configuration...
[OK]
ASW1#show port-security
Secure Port MaxSecureAddr CurrentAddr SecurityViolation Security Action
               (Count)       (Count)         (Count)
--------------------------------------------------------------------------
        Fa0/2         2           0                0         Restrict
        Fa0/3         2           0                0         Restrict
        Fa0/4         2           0                0         Restrict
        Fa0/5         2           0                0         Restrict
--------------------------------------------------------------------------
ASW1#
```

Configure DHCP Snooping on ASW1

```
---------------------------------------------------------------
ASW1#enable
ASW1#configure terminal
Enter configuration commands, one per line.  End with CNTL/Z.
ASW1(config)#ip dhcp snooping
ASW1(config)#ip dhcp snooping vlan 30,60
ASW1(config)#interface fastEthernet 0/1
ASW1(config-if)#ip dhcp snooping trust
ASW1(config-if)#end
ASW1#
%SYS-5-CONFIG_I: Configured from console by console

ASW1#wr
Building configuration...
[OK]
ASW1#show ip dhcp snooping
Switch DHCP snooping is enabled
DHCP snooping is configured on following VLANs:
30,60
Insertion of option 82 is enabled
Option 82 on untrusted port is not allowed
Verification of hwaddr field is enabled
Interface                  Trusted    Rate limit (pps)
-----------------------    -------    ----------------
FastEthernet0/1            yes        unlimited
ASW1#
```

Dynamic ARP Inspection Configuration on ASW1

```
Total number of bindings: 1
ASW1#enable
ASW1#configure terminal
Enter configuration commands, one per line.  End with CNTL/Z.
ASW1(config)#ip arp inspection vlan 30
ASW1(config)#interface fastEthernet 0/1
ASW1(config-if)#ip arp inspection trust
ASW1(config-if)#end
ASW1#
```

Administrative shutdown of all unused ports, assigned to the parking VLAN on ASW1.

```
ASW1(config)#
ASW1(config)#interface range fastEthernet 0/6 - 24
ASW1(config-if-range)#switchport mode access
ASW1(config-if-range)#switchport access vlan 999
ASW1(config-if-range)#shutdown
ASW1(config-if-range)#exit
ASW1(config)#
ASW1(config)#interface range gigabitEthernet 0/1 - 2
ASW1(config-if-range)#switchport mode access
ASW1(config-if-range)#switchport access vlan 999
ASW1(config-if-range)#shutdown

%LINK-5-CHANGED: Interface GigabitEthernet0/1, changed state to
administratively down

%LINK-5-CHANGED: Interface GigabitEthernet0/2, changed state to
administratively down
ASW1(config-if-range)#exit
ASW1(config)#
ASW1(config)#end
ASW1#write memory
%SYS-5-CONFIG_I: Configured from console by console

ASW1#write memory
Building configuration...
```

Configure Port Security on ASW2

```
ASW2#
ASW2#configure terminal
Enter configuration commands, one per line.  End with CNTL/Z.
ASW2(config)#
ASW2(config)#interface range fastEthernet 0/2 - 3
ASW2(config-if-range)#switchport mode access
ASW2(config-if-range)#switchport port-security
ASW2(config-if-range)#switchport port-security maximum 2
ASW2(config-if-range)#switchport port-security mac-address sticky
ASW2(config-if-range)#switchport port-security violation restrict
ASW2(config-if-range)#exit
ASW2(config)#
ASW2(config)#end
ASW2#write memory
%SYS-5-CONFIG_I: Configured from console by console

ASW2#write memory
Building configuration...
[OK]
ASW2#show port-security
Secure Port MaxSecureAddr CurrentAddr SecurityViolation Security Action
               (Count)       (Count)        (Count)
---------------------------------------------------------------------
        Fa0/2        2           0                0         Restrict
        Fa0/3        2           0                0         Restrict
---------------------------------------------------------------------
ASW2#
```

Configured DHCP Snooping on ASW2

```
ASW2#configure terminal
Enter configuration commands, one per line.  End with CNTL/Z.
ASW2(config)#
ASW2(config)#ip dhcp snooping
ASW2(config)#ip dhcp snooping vlan 20
ASW2(config)#no ip dhcp snooping information option
ASW2(config)#
ASW2(config)#interface fastEthernet 0/1
ASW2(config-if)#ip dhcp snooping trust
ASW2(config-if)#exit
ASW2(config)#
ASW2(config)#end
ASW2#write memory
%SYS-5-CONFIG_I: Configured from console by console

ASW2#write memory
Building configuration...
[OK]
ASW2#show ip dhcp snooping
Switch DHCP snooping is enabled
DHCP snooping is configured on following VLANs:
20
Insertion of option 82 is disabled
Option 82 on untrusted port is not allowed
Verification of hwaddr field is enabled
Interface                  Trusted    Rate limit (pps)
-----------------------    -------    ----------------
FastEthernet0/1            yes        unlimited
ASW2#
```

Configured Dynamic ARP Inspection on ASW2

```
ASW2#configure terminal
Enter configuration commands, one per line.  End with CNTL/Z.
ASW2(config)#
ASW2(config)#ip arp inspection vlan 20
ASW2(config)#
ASW2(config)#interface fastEthernet 0/1
ASW2(config-if)#ip arp inspection trust
ASW2(config-if)#exit
ASW2(config)#
ASW2(config)#end
ASW2#write memory
%SYS-5-CONFIG_I: Configured from console by console

ASW2#write memory
Building configuration...
```

Administrative shutdown of all unused ports, assigned to the parking VLAN on ASW2

```
Enter configuration commands, one per line.  End with CNTL/Z.
ASW2(config)#
ASW2(config)#interface range fastEthernet 0/4 - 24
ASW2(config-if-range)#switchport mode access
ASW2(config-if-range)#switchport access vlan 999
ASW2(config-if-range)#shutdown
```

```
%LINK-5-CHANGED: Interface FastEthernet0/24, changed state to administratively down
ASW2(config-if-range)#exit
ASW2(config)#
ASW2(config)#interface range gigabitEthernet 0/1 - 2
ASW2(config-if-range)#switchport mode access
ASW2(config-if-range)#switchport access vlan 999
ASW2(config-if-range)#shutdown

%LINK-5-CHANGED: Interface GigabitEthernet0/1, changed state to administratively down

%LINK-5-CHANGED: Interface GigabitEthernet0/2, changed state to administratively down
ASW2(config-if-range)#exit
ASW2(config)#
ASW2(config)#end
ASW2#write memory
%SYS-5-CONFIG_I: Configured from console by console

ASW2#write memory
Building configuration...
[OK]
```

Configure Port Security on ASW3

```
ASW3#configure terminal
Enter configuration commands, one per line.  End with CNTL/Z.
ASW3(config)#
ASW3(config)#interface range fastEthernet 0/2 - 3
ASW3(config-if-range)#switchport mode access
ASW3(config-if-range)#switchport port-security
ASW3(config-if-range)#switchport port-security maximum 2
ASW3(config-if-range)#switchport port-security mac-address sticky
ASW3(config-if-range)#switchport port-security violation restrict
ASW3(config-if-range)#exit
ASW3(config)#
ASW3(config)#end
ASW3#write memory
%SYS-5-CONFIG_I: Configured from console by console

ASW3#write memory
Building configuration...
[OK]
```

DHCP Snooping on ASW3

```
ASW3#configure terminal
Enter configuration commands, one per line.  End with CNTL/Z.
ASW3(config)#
ASW3(config)#ip dhcp snooping
ASW3(config)#ip dhcp snooping vlan 70
ASW3(config)#no ip dhcp snooping information option
ASW3(config)#
ASW3(config)#interface fastEthernet 0/1
ASW3(config-if)#ip dhcp snooping trust
ASW3(config-if)#exit
ASW3(config)#
ASW3(config)#end
ASW3#write memory
%SYS-5-CONFIG_I: Configured from console by console

ASW3#write memory
Building configuration...
[OK]
```

Dynamic ARP Inspection on ASW3

```
FastEthernet0/1            yes         unlimited
ASW3#configure terminal
Enter configuration commands, one per line.  End with CNTL/Z.
ASW3(config)#
ASW3(config)#ip arp inspection vlan 70
ASW3(config)#
ASW3(config)#interface fastEthernet 0/1
ASW3(config-if)#ip arp inspection trust
ASW3(config-if)#exit
ASW3(config)#
ASW3(config)#end
ASW3#write memory
%SYS-5-CONFIG_I: Configured from console by console
```

Administrative shutdown of all unused ports, assigned to the parking VLAN on ASW3

```
Total number of bindings: 1
ASW3#configure terminal
Enter configuration commands, one per line.  End with CNTL/Z.
ASW3(config)#
ASW3(config)#interface range fastEthernet 0/5 - 24
ASW3(config-if-range)#switchport mode access
ASW3(config-if-range)#switchport access vlan 999
ASW3(config-if-range)#shutdown

ASW3(config-if-range)#exit
ASW3(config)#
ASW3(config)#interface range gigabitEthernet 0/1 - 2
ASW3(config-if-range)#switchport mode access
ASW3(config-if-range)#switchport access vlan 999
ASW3(config-if-range)#shutdown

%LINK-5-CHANGED: Interface GigabitEthernet0/1, changed state to administratively down

%LINK-5-CHANGED: Interface GigabitEthernet0/2, changed state to administratively down
ASW3(config-if-range)#exit
ASW3(config)#
ASW3(config)#end
ASW3#write memory
%SYS-5-CONFIG_I: Configured from console by console

ASW3#write memory
Building configuration...
[OK]
ASW3#wr
Building configuration...
[OK]
ASW3#
```

Configure Port Security on ASW4

```
ASW4#configure terminal
Enter configuration commands, one per line.  End with CNTL/Z.
ASW4(config)#
ASW4(config)#interface range fastEthernet 0/2 - 3
ASW4(config-if-range)#switchport mode access
ASW4(config-if-range)#switchport port-security
ASW4(config-if-range)#switchport port-security maximum 2
ASW4(config-if-range)#switchport port-security mac-address sticky
ASW4(config-if-range)#switchport port-security violation restrict
ASW4(config-if-range)#exit
ASW4(config)#
ASW4(config)#end
ASW4#write memory
%SYS-5-CONFIG_I: Configured from console by console

ASW4#write memory
Building configuration...
[OK]
```

DHCP Snooping on ASW4

```
ASW4>en
Password:
ASW4#configure terminal
Enter configuration commands, one per line.  End with CNTL/Z.
ASW4(config)#
ASW4(config)#ip dhcp snooping
ASW4(config)#ip dhcp snooping vlan 80
ASW4(config)#no ip dhcp snooping information option
ASW4(config)#
ASW4(config)#interface fastEthernet 0/1
ASW4(config-if)#ip dhcp snooping trust
ASW4(config-if)#exit
ASW4(config)#
ASW4(config)#end
ASW4#write memory
%SYS-5-CONFIG_I: Configured from console by console

ASW4#write memory
Building configuration...
```

Dynamic ARP Inspection on ASW4

```
ASW4#configure terminal
Enter configuration commands, one per line.  End with CNTL/Z.
ASW4(config)#
ASW4(config)#ip arp inspection vlan 80
ASW4(config)#
ASW4(config)#interface fastEthernet 0/1
ASW4(config-if)#ip arp inspection trust
ASW4(config-if)#exit
ASW4(config)#
ASW4(config)#end
ASW4#write memory
%SYS-5-CONFIG_I: Configured from console by console

ASW4#write memory
```

Administrative shutdown of all unused ports, assigned to the parking VLAN on ASW4

```
Enter configuration commands, one per line.  End with CNTL/Z.
ASW4(config)#
ASW4(config)#interface range fastEthernet 0/5 - 24
ASW4(config-if-range)#switchport mode access
ASW4(config-if-range)#switchport access vlan 999
ASW4(config-if-range)#shutdown


%LINK-5-CHANGED: Interface FastEthernet0/24, changed state to administratively down
ASW4(config-if-range)#exit
ASW4(config)#
ASW4(config)#interface range gigabitEthernet 0/1 - 2
ASW4(config-if-range)#switchport mode access
ASW4(config-if-range)#switchport access vlan 999
ASW4(config-if-range)#shutdown

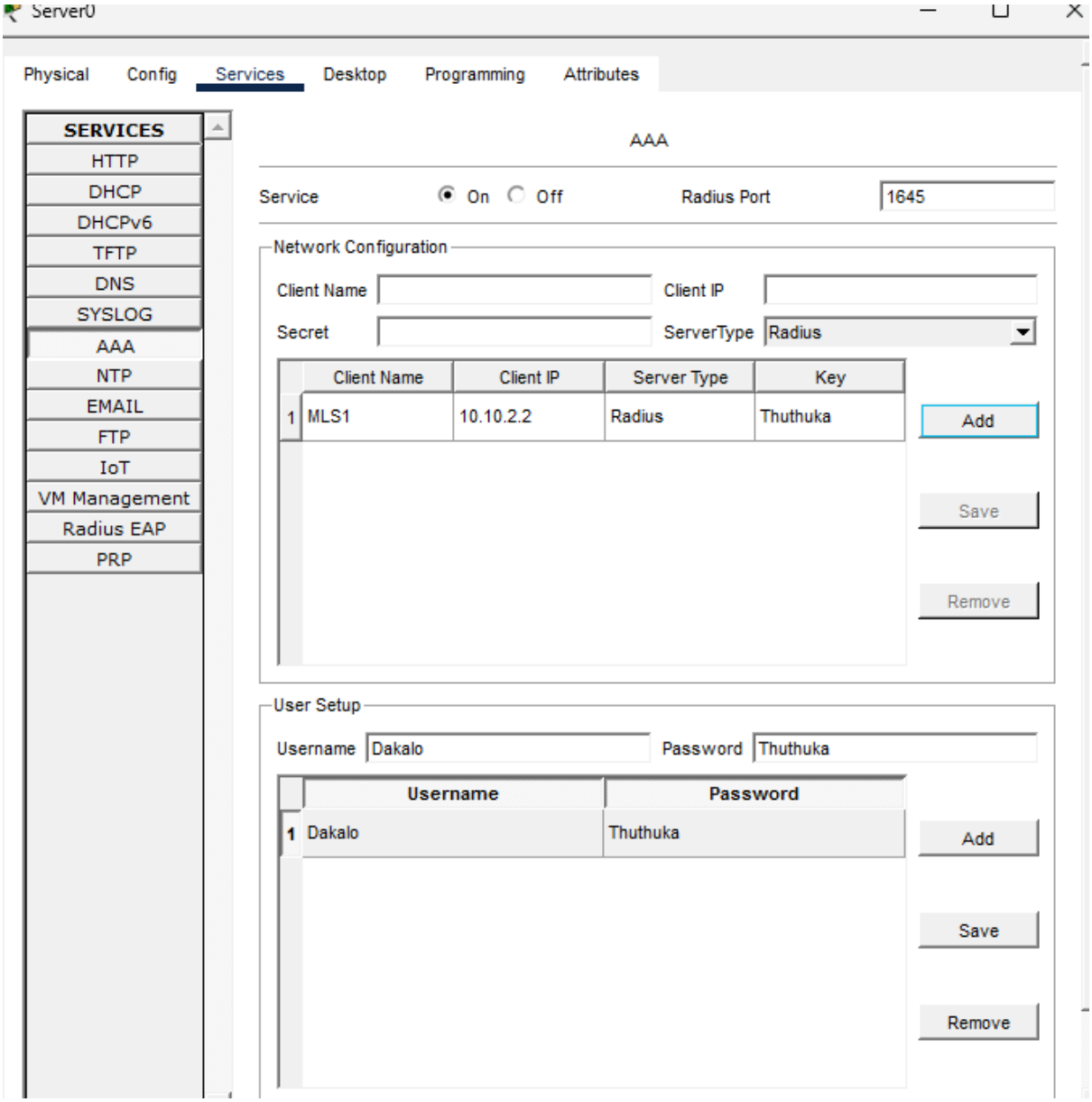
%LINK-5-CHANGED: Interface GigabitEthernet0/1, changed state to administratively down

%LINK-5-CHANGED: Interface GigabitEthernet0/2, changed state to administratively down
ASW4(config-if-range)#exit
ASW4(config)#
ASW4(config)#end
ASW4#write memory
%SYS-5-CONFIG_I: Configured from console by console

ASW4#write memory
Building configuration...
[OK]
```

3.4 Management Plane Security and Centralised Services

RADIUS authentication was enabled on Server0 (10.10.2.10). MLS1 (10.10.2.2) was registered as a RADIUS client using a shared secret, and an administrator account was created for centralised network authentication.



AAA was enabled on MLS1, a local administrator account was created for fallback authentication, and the RADIUS server (10.10.2.10) was configured for centralised authentication.

```
MLS1#configure terminal
Enter configuration commands, one per line.  End with CNTL/Z.
MLS1(config)#
MLS1(config)#username localadmin privilege 15 secret LocalAdmin2026
MLS1(config)#aaa new-model
MLS1(config)#
MLS1(config)#radius-server host 10.10.2.10 auth-port 1645 key Thuthuka
MLS1(config)#
MLS1(config)#end
MLS1#write memory
%SYS-5-CONFIG_I: Configured from console by console
```

MLS1 was configured to use RADIUS authentication as the primary login method, with a local administrator account as a fallback if the RADIUS server becomes unavailable.

```
MLS1>en
Password:
MLS1#configure terminal
Enter configuration commands, one per line.  End with CNTL/Z.
MLS1(config)#
MLS1(config)#aaa authentication login default group radius local
MLS1(config)#
MLS1(config)#end
MLS1#write memory
%SYS-5-CONFIG_I: Configured from console by console

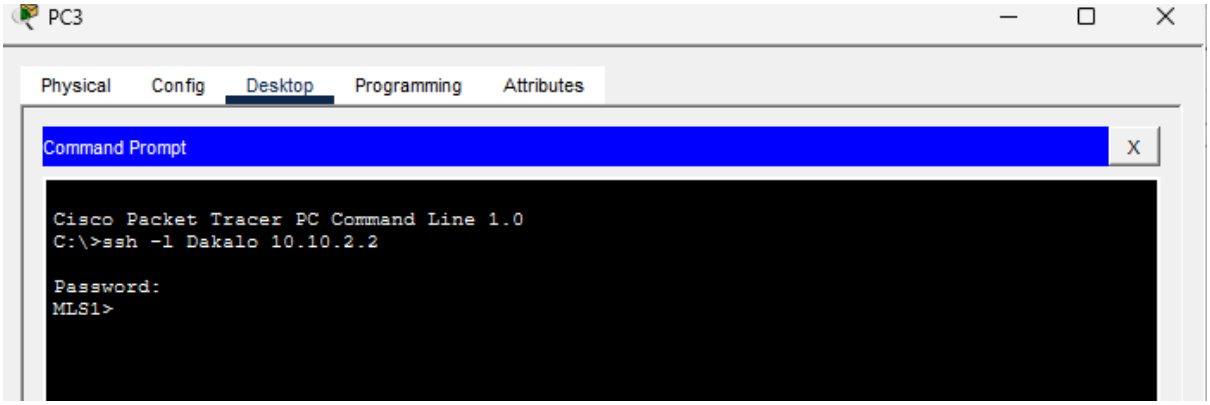
MLS1#write memory
Building configuration...
[OK]
MLS1#show running-config | include aaa
aaa new-model
aaa authentication login default group radius local
MLS1#
```

MLS1 was configured to use AAA authentication on all VTY lines, allowing secure SSH access with RADIUS authentication and local fallback. Telnet access was disabled, and a five-minute inactivity timeout was maintained.

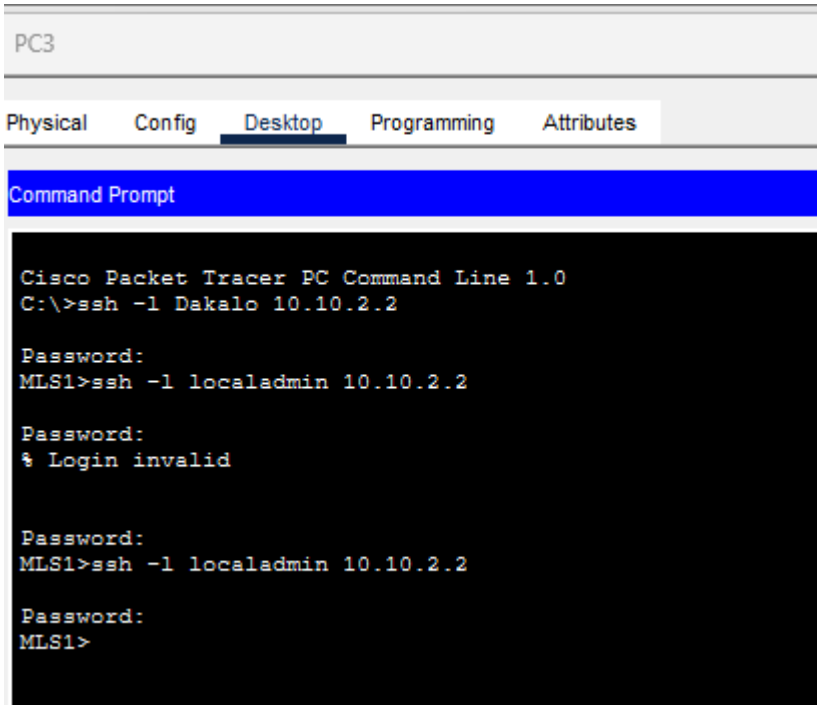
```
 transport input ssh
MLS1#configure terminal
Enter configuration commands, one per line.  End with CNTL/Z.
MLS1(config)#
MLS1(config)#line vty 0 15
MLS1(config-line)#login authentication default
MLS1(config-line)#transport input ssh
MLS1(config-line)#exit
MLS1(config)#
MLS1(config)#end
MLS1#write memory
%SYS-5-CONFIG_I: Configured from console by console

MLS1#write memory
Building configuration...
[OK]
MLS1#show running-config | section line vty
line vty 0 4
 exec-timeout 5 0
 login authentication default
 transport input ssh
line vty 5 15
 exec-timeout 5 0
 login authentication default
 transport input ssh
MLS1#
```

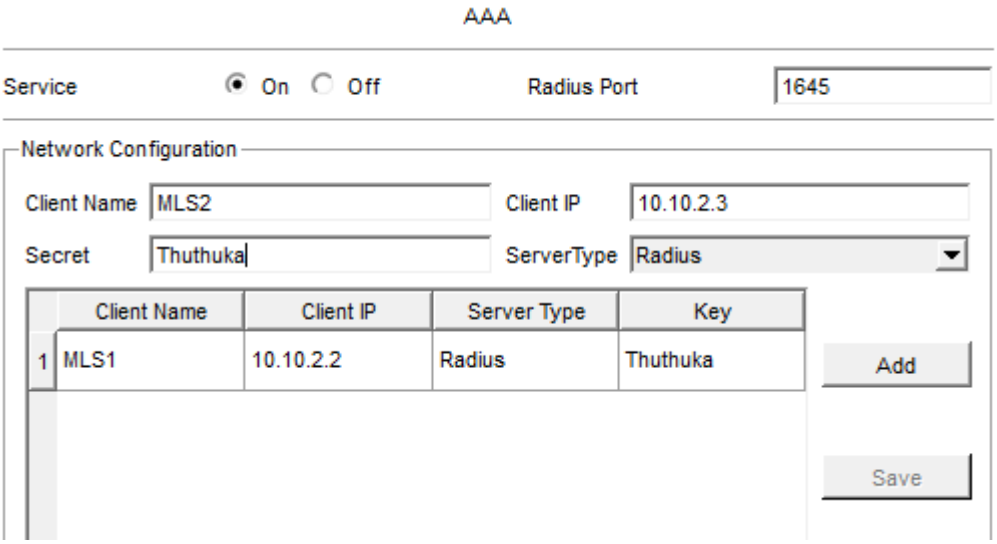
C3 successfully connected to MLS1 using SSH with the RADIUS user account Dakalo. The MLS1> prompt confirms that the login was successful, demonstrating remote access through the configured AAA authentication method.



The local administrator account successfully logged into MLS1 through SSH while the RADIUS service was unavailable. This demonstrates that local fallback authentication allows administrative access when the central authentication server is offline.



MLS2 was registered on Server0 as a RADIUS client using IP address 10.10.2.3 and a shared secret, allowing it to use centralised authentication services.

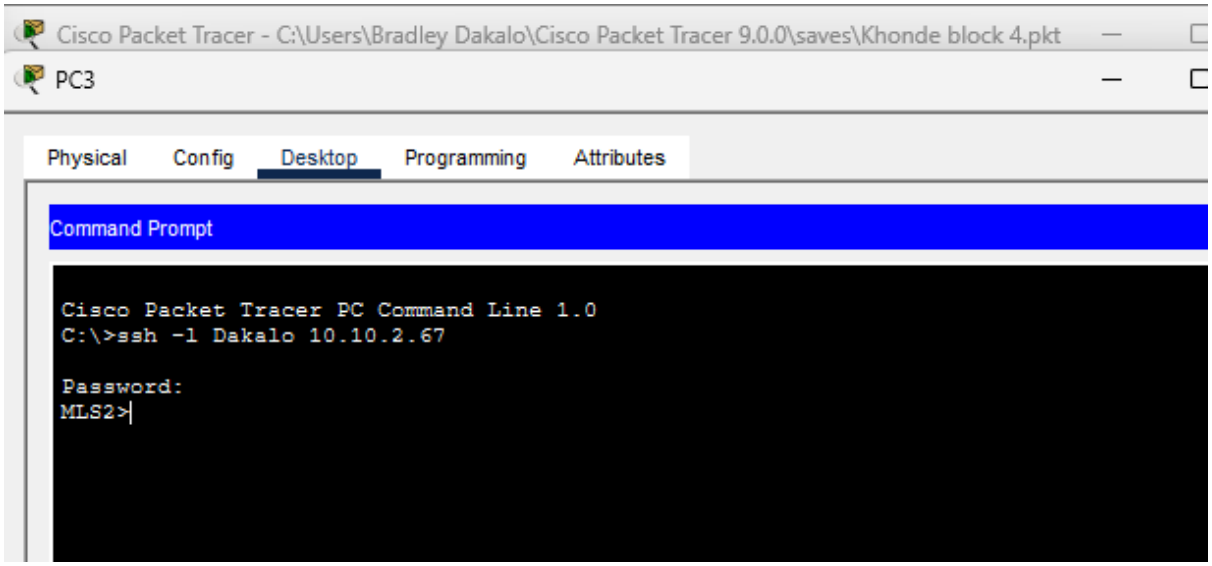


MLS2 was configured to use RADIUS authentication with a local administrator account as fallback. AAA was enabled, and SSH authentication was applied to the VTY lines.

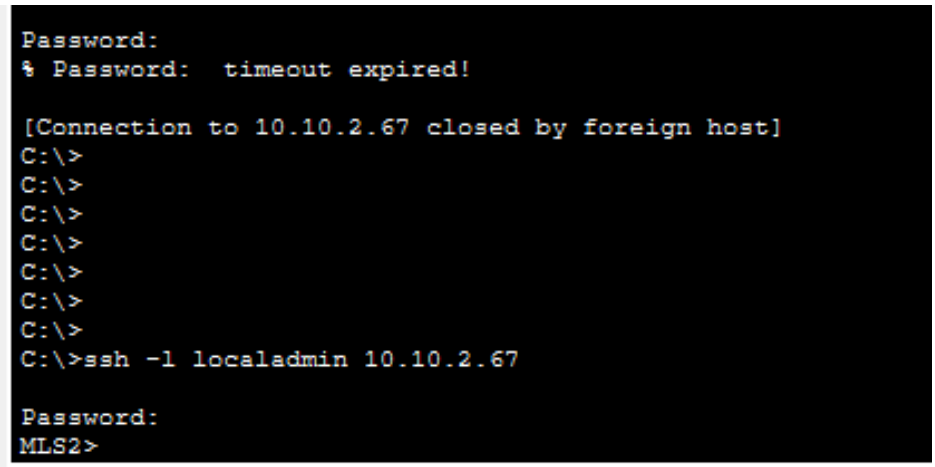
```
LS2#configure terminal
nter configuration commands, one per line.  End with CNTL/Z.
LS2(config)#
LS2(config)#username localadmin privilege 15 secret Thuthuka
LS2(config)#aaa new-model
LS2(config)#
LS2(config)#radius-server host 10.10.2.10 auth-port 1645 key Thuthuka
LS2(config)#ip radius source-interface Vlan10
                 ^
 Invalid input detected at '^' marker.

LS2(config)#
LS2(config)#aaa authentication login default group radius local
LS2(config)#
LS2(config)#line vty 0 15
LS2(config-line)#login authentication default
LS2(config-line)#transport input ssh
LS2(config-line)#exit
LS2(config)#
LS2(config)#end
LS2#write memory
SYS-5-CONFIG_I: Configured from console by console
```

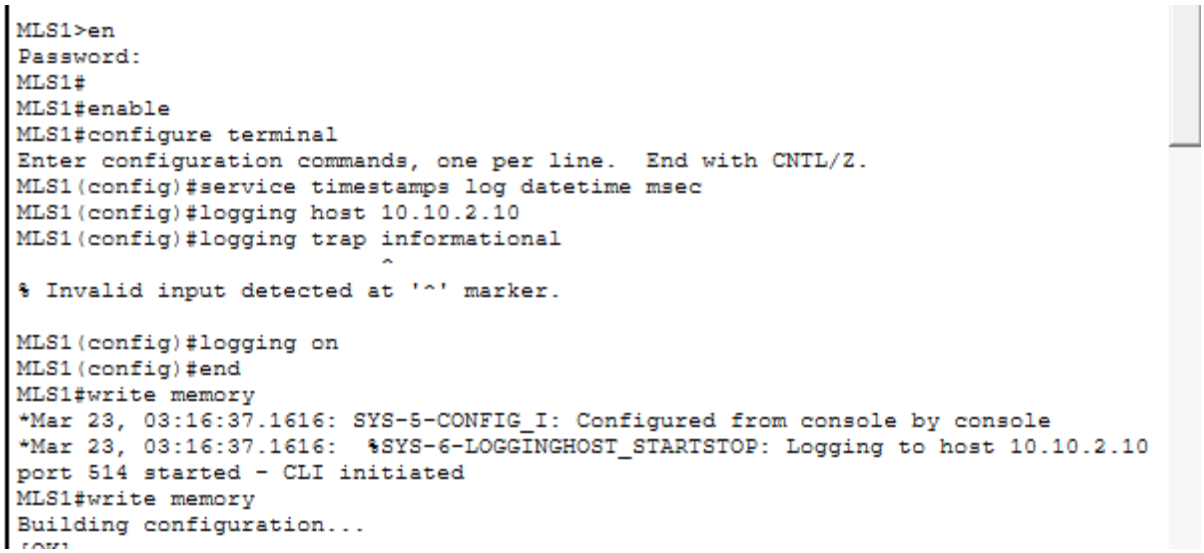
The screenshot shows successful SSH authentication from PC3 to MLS2 using the RADIUS username Dakalo and management IP address 10.10.2.67. The MLS2> prompt confirms that remote login was successful, demonstrating that centralised AAA authentication through the RADIUS server is functioning correctly.



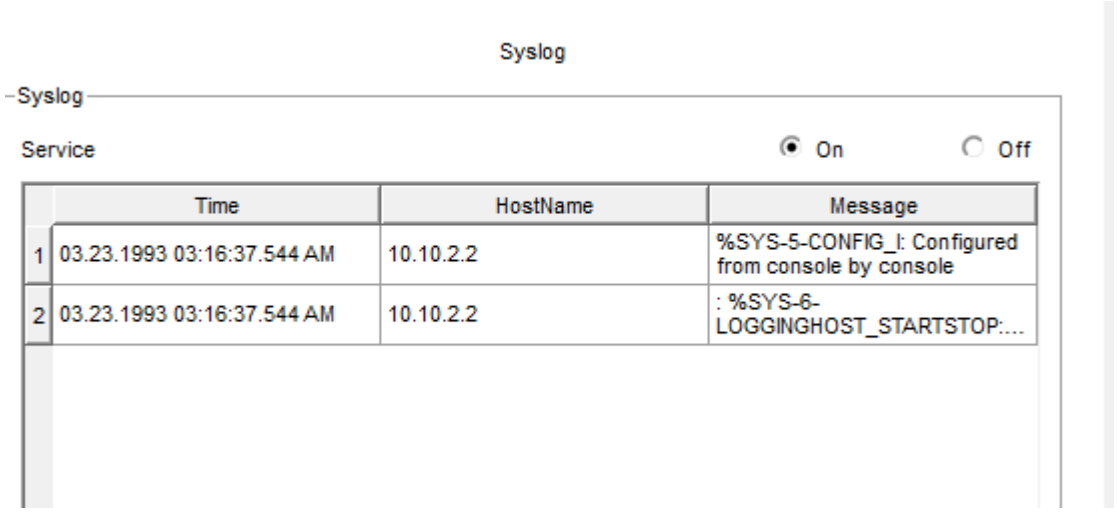
shows successful SSH login from PC3 to MLS2 using the local administrator account (localadmin) at IP address 10.10.2.67. The MLS2> prompt confirms that local fallback authentication works when the RADIUS server is unavailable, ensuring administrators can still access the switch.



shows MLS1 configured to send Syslog messages to Server0 (10.10.2.10) using UDP port 514. Logging was enabled and the configuration was saved successfully.



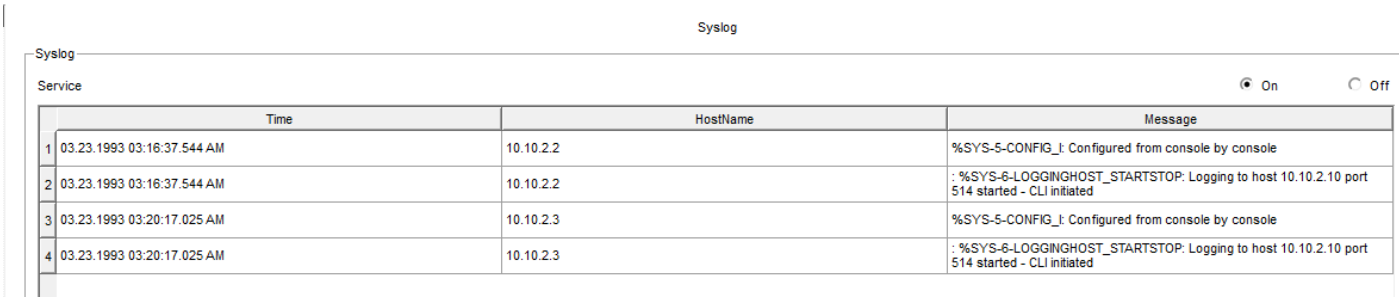
Confirms that Server0 successfully receives and records Syslog messages from MLS1 (10.10.2.2), demonstrating centralised logging.



confirms that MLS2 is configured to send Syslog messages to Server0 (10.10.2.10) using UDP port 514 at informational severity level.

```
Username: Dakalo
Password:
MLS2>en
Password:
MLS2#configure terminal
Enter configuration commands, one per line.  End with CNTL/Z.
MLS2(config)#service timestamps log datetime msec
MLS2(config)#logging host 10.10.2.10
MLS2(config)#logging on
MLS2(config)#end
MLS2#write memory
*Mar 23, 03:20:17.2020: SYS-5-CONFIG_I: Configured from console by console
*Mar 23, 03:20:17.2020:  %SYS-6-LOGGINGHOST_STARTSTOP: Logging to host 10.10.2.10 port
514 started - CLI initiated
MLS2#write memory
Building configuration...
[OK]
MLS2#
```

confirms that Server0 successfully receives and records centralised Syslog messages from both MLS1 and MLS2.



shows ASW1 successfully configured to send Syslog messages to Server0 (10.10.2.10) using UDP port 514 at informational severity level.

```
ASW1>en
Password:
ASW1#configure terminal
Enter configuration commands, one per line.  End with CNTL/Z.
ASW1(config)#service timestamps log datetime msec
ASW1(config)#logging host 10.10.2.10
ASW1(config)#logging on
ASW1(config)#end
ASW1#write memory
*Mar 23, 03:24:37.2424: SYS-5-CONFIG_I: Configured from console by console
*Mar 23, 03:24:37.2424:  %SYS-6-LOGGINGHOST_STARTSTOP: Logging to host 10.10.2.10 port 514 started - CLI
initiated
ASW1#write memory
Building configuration...
[OK]
```

Configure Syslog on ASW2

```
ASW2#configure terminal
Enter configuration commands, one per line.  End with CNTL/Z.
ASW2(config)#service timestamps log datetime msec
ASW2(config)#logging host 10.10.2.10
ASW2(config)#logging on
ASW2(config)#end
ASW2#write memory
*Mar 23, 03:37:32.3737: SYS-5-CONFIG_I: Configured from console by console
*Mar 23, 03:37:32.3737:  %SYS-6-LOGGINGHOST_STARTSTOP: Logging to host 10.10.2.10 port 514
started - CLI initiated
ASW2#write memory
Building configuration...
```

Configure Syslog on ASW3

```
ASW3#configure terminal
Enter configuration commands, one per line.  End with CNTL/Z.
ASW3(config)#service timestamps log datetime msec
ASW3(config)#logging host 10.10.2.10
ASW3(config)#logging on
ASW3(config)#end
ASW3#write memory
*Mar 23, 03:42:39.4242: SYS-5-CONFIG_I: Configured from console by console
*Mar 23, 03:42:39.4242:  %SYS-6-LOGGINGHOST_STARTSTOP: Logging to host 10.10.2.10 port 514 started - CLI initiated
ASW3#write memory
Building configuration...
[OK]
ASW3#
ASW3#
```

Configure Syslog on ASW4

```
ASW4#configure terminal
Enter configuration commands, one per line.  End with CNTL/Z.
ASW4(config)#service timestamps log datetime msec
ASW4(config)#logging host 10.10.2.10
ASW4(config)#logging on
ASW4(config)#end
ASW4#write memory
*Mar 23, 03:46:52.4646: SYS-5-CONFIG_I: Configured from console by console
*Mar 23, 03:46:52.4646:  %SYS-6-LOGGINGHOST_STARTSTOP: Logging to host 10.10.2.10 port 514 started - CLI initiated
ASW4#write memory
Building configuration...
[OK]
ASW4#
ASW4#
```

Configure Syslog on R1

```
R1#
R1#configure terminal
Enter configuration commands, one per line.  End with CNTL/Z.
R1(config)#service timestamps log datetime msec
R1(config)#logging host 10.10.2.10
R1(config)#logging trap informational
                  ^
% Invalid input detected at '^' marker.

R1(config)#logging on
R1(config)#end
R1#write memory
*Mar 23, 03:11:05.1111: SYS-5-CONFIG_I: Configured from console by console
*Mar 23, 03:11:05.1111:  %SYS-6-LOGGINGHOST_STARTSTOP: Logging to host 10.10.2.10 port 514 started - CLI initiated
R1#write memory
Building configuration...
[OK]
R1#
R1#wr
Building configuration...
[OK]
R1#
```

Configure centralised Syslog on R2

```
R2#
R2#
R2#
R2#configure terminal
Enter configuration commands, one per line.  End with CNTL/Z.
R2(config)#service timestamps log datetime msec
R2(config)#logging host 10.10.2.10
R2(config)#logging trap informational
                  ^
% Invalid input detected at '^' marker.

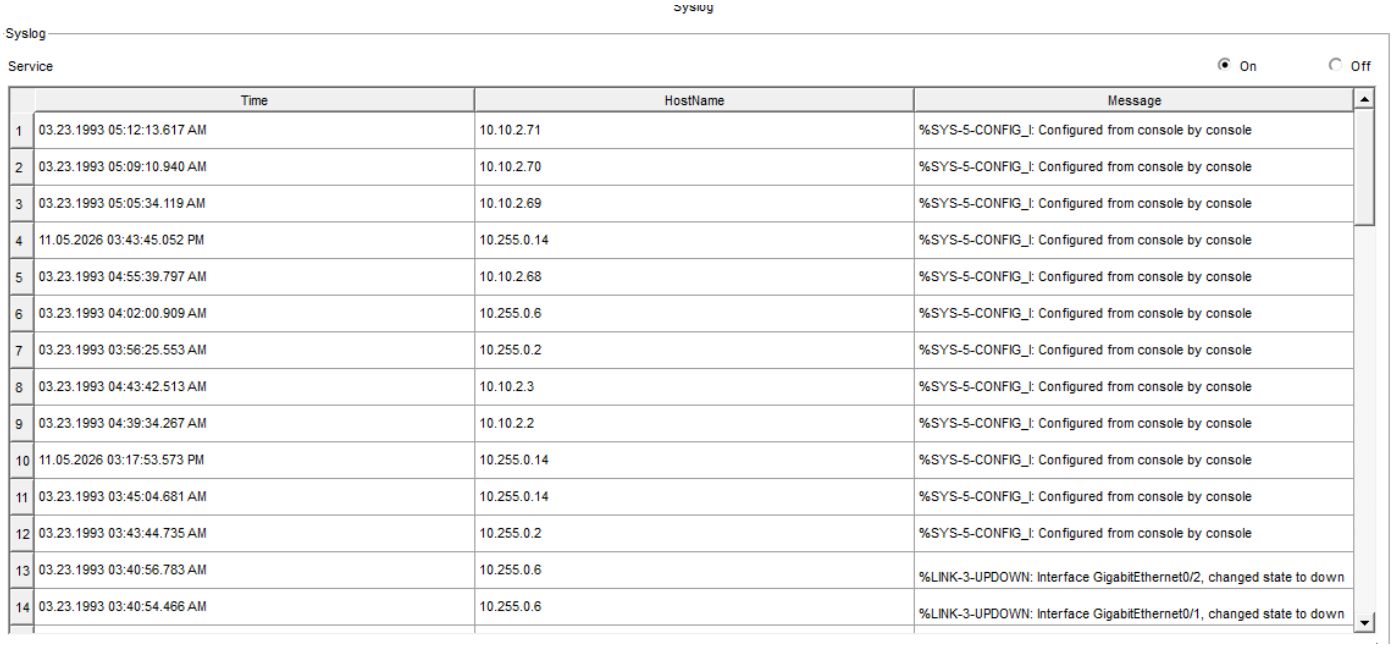
R2(config)#logging on
R2(config)#end
R2#write memory
*Mar 23, 03:25:57.2525: SYS-5-CONFIG_I: Configured from console by console
*Mar 23, 03:25:57.2525:  %SYS-6-LOGGINGHOST_STARTSTOP: Logging to host 10.10.2.10 port 514 started - CLI initiated
R2#write memory
Building configuration...
[OK]
R2#
```

Configure R3 centralised Syslog

```
R3#configure terminal
Enter configuration commands, one per line.  End with CNTL/Z.
R3(config)#service timestamps log datetime msec
R3(config)#logging host 10.10.2.10
R3(config)#logging trap informational
                  ^
% Invalid input detected at '^' marker.

R3(config)#logging on
R3(config)#end
R3#write memory
*Mar 23, 03:35:30.3535: SYS-5-CONFIG_I: Configured from console by console
*Mar 23, 03:35:30.3535:  %SYS-6-LOGGINGHOST_STARTSTOP: Logging to host 10.10.2.10 port 514 started - CLI initiated
R3#write memory
Building configuration...
[OK]
R3#
```

Syslog messages collected from the network routers and switches on Server0 (10.10.2.10), including configuration changes and interface status events.



	Time	HostName	Message
1	03.23.1993 05:12:13.617 AM	10.10.2.71	%SYS-5-CONFIG_I: Configured from console by console
2	03.23.1993 05:09:10.940 AM	10.10.2.70	%SYS-5-CONFIG_I: Configured from console by console
3	03.23.1993 05:05:34.119 AM	10.10.2.69	%SYS-5-CONFIG_I: Configured from console by console
4	11.05.2026 03:43:45.052 PM	10.255.0.14	%SYS-5-CONFIG_I: Configured from console by console
5	03.23.1993 04:55:39.797 AM	10.10.2.68	%SYS-5-CONFIG_I: Configured from console by console
6	03.23.1993 04:02:00.909 AM	10.255.0.6	%SYS-5-CONFIG_I: Configured from console by console
7	03.23.1993 03:56:25.553 AM	10.255.0.2	%SYS-5-CONFIG_I: Configured from console by console
8	03.23.1993 04:43:42.513 AM	10.10.2.3	%SYS-5-CONFIG_I: Configured from console by console
9	03.23.1993 04:39:34.267 AM	10.10.2.2	%SYS-5-CONFIG_I: Configured from console by console
10	11.05.2026 03:17:53.573 PM	10.255.0.14	%SYS-5-CONFIG_I: Configured from console by console
11	03.23.1993 03:45:04.681 AM	10.255.0.14	%SYS-5-CONFIG_I: Configured from console by console
12	03.23.1993 03:43:44.735 AM	10.255.0.2	%SYS-5-CONFIG_I: Configured from console by console
13	03.23.1993 03:40:56.783 AM	10.255.0.6	%LINK-3-UPDOWN: Interface GigabitEthernet0/2, changed state to down
14	03.23.1993 03:40:54.466 AM	10.255.0.6	%LINK-3-UPDOWN: Interface GigabitEthernet0/1, changed state to down

	Time	HostName	Message
47	03.23.1993 03:46:52.030 AM	10.10.2.71	: %SYS-6-LOGGINGHOST_STARTSTOP: Logging to host 10.10.2.10 port 514 started - CLI initiated
48	03.23.1993 03:46:52.030 AM	10.10.2.71	%SYS-5-CONFIG_I: Configured from console by console
49	03.23.1993 03:44:12.460 AM	10.10.2.70	%SYS-5-CONFIG_I: Configured from console by console
50	03.23.1993 03:42:39.430 AM	10.10.2.70	: %SYS-6-LOGGINGHOST_STARTSTOP: Logging to host 10.10.2.10 port 514 started - CLI initiated
51	03.23.1993 03:42:39.430 AM	10.10.2.70	%SYS-5-CONFIG_I: Configured from console by console
52	03.23.1993 03:16:37.544 AM	10.10.2.2	%SYS-5-CONFIG_I: Configured from console by console
53	03.23.1993 03:16:37.544 AM	10.10.2.2	: %SYS-6-LOGGINGHOST_STARTSTOP: Logging to host 10.10.2.10 port 514 started - CLI initiated
54	03.23.1993 03:20:17.025 AM	10.10.2.3	%SYS-5-CONFIG_I: Configured from console by console
55	03.23.1993 03:20:17.025 AM	10.10.2.3	: %SYS-6-LOGGINGHOST_STARTSTOP: Logging to host 10.10.2.10 port 514 started - CLI initiated
56	03.23.1993 03:31:44.957 AM	10.10.2.68	%SYS-5-CONFIG_I: Configured from console by console
57	03.23.1993 03:37:32.053 AM	10.10.2.69	%SYS-5-CONFIG_I: Configured from console by console
58	03.23.1993 03:37:32.053 AM	10.10.2.69	: %SYS-6-LOGGINGHOST_STARTSTOP: Logging to host 10.10.2.10 port 514 started - CLI initiated
59	03.23.1993 03:40:24.037 AM	10.10.2.69	%SYS-5-CONFIG_I: Configured from console by console
60	03.23.1993 03:40:42.012 AM	10.10.2.69	%SYS-5-CONFIG_I: Configured from console by console

R1 is configured as the authoritative NTP server using MD5 authentication. The output confirms successful clock synchronisation and enabled NTP authentication.

```
R1#
R1#configure terminal
Enter configuration commands, one per line.  End with CNTL/Z.
R1(config)#ntp authentication-key 1 md5 Thuthuka
R1(config)#ntp authenticate
R1(config)#ntp trusted-key 1
R1(config)#ntp master 3
R1(config)#end
R1#write memory
*Nov 05, 15:17:53.1717: SYS-5-CONFIG_I: Configured from console by console
R1#write memory
Building configuration...
[OK]
R1#show ntp status
Clock is synchronized, stratum 1, reference is 127.127.1.1
nominal freq is 250.0000 Hz, actual freq is 249.9990 Hz, precision is 2**24
reference time is EE68F821.0000023C (15:17:53.572 UTC Thu Nov 5 2026)
clock offset is 0.00 msec, root delay is 0.00  msec
root dispersion is 0.00 msec, peer dispersion is 0.01 msec.
loopfilter state is 'CTRL' (Normal Controlled Loop), drift is - 0.000001193 s/s system
poll interval is 4, last update was 8 sec ago.
R1#show running-config | include ntp
ntp authentication-key 1 md5 0815445B1D11101C13 7
ntp authenticate
ntp trusted-key 1
ntp master 3
R1#
```

Configure MLS1 as an authenticated NTP client

```
Username: Dakalo
Password:
MLS1>en
Password:
MLS1#configure terminal
Enter configuration commands, one per line.  End with CNTL/Z.
MLS1(config)#ntp authentication-key 1 md5 Thuthuka
MLS1(config)#ntp authenticate
MLS1(config)#ntp trusted-key 1
MLS1(config)#ntp server 10.255.0.14 key 1
MLS1(config)#end
MLS1#
*Mar 23, 04:39:34.3939: SYS-5-CONFIG_I: Configured from console by console
MLS1#wr
Building configuration...
[OK]
```

Configure MLS2 as an NTP client

```
Unauthorised access is not allowed.

User Access Verification

Username: Dakalo
Password:
MLS2>en
Password:
MLS2#configure terminal
Enter configuration commands, one per line.  End with CNTL/Z.
MLS2(config)#ntp authentication-key 1 md5 Thuthuka
MLS2(config)#ntp authenticate
MLS2(config)#ntp trusted-key 1
MLS2(config)#ntp server 10.255.0.18 key 1
MLS2(config)#end
MLS2#
*Mar 23, 04:43:42.4343: SYS-5-CONFIG_I: Configured from console by console
MLS2#
MLS2#
```

R2 Authenticated NTP Client Configuration

```
Password:

R2>en
Password:
R2#configure terminal
Enter configuration commands, one per line.  End with CNTL/Z.
R2(config)#ntp authentication-key 1 md5 Thuthuka
R2(config)#ntp authenticate
R2(config)#ntp trusted-key 1
R2(config)#ntp server 10.255.0.1 key 1
R2(config)#end
R2#
*Mar 23, 03:56:25.5656: SYS-5-CONFIG_I: Configured from console by console
R2#wr
Building configuration...
[OK]
R2#
```

R3 Authenticated NTP Client Configuration

```
Password:

R3>en
Password:
R3#configure terminal
Enter configuration commands, one per line.  End with CNTL/Z.
R3(config)#ntp authentication-key 1 md5 Thuthuka
R3(config)#ntp authenticate
R3(config)#ntp trusted-key 1
R3(config)#ntp server 10.255.0.5 key 1
R3(config)#end
R3#write memory
*Mar 23, 04:02:00.022: SYS-5-CONFIG_I: Configured from console by
console
R3#write memory
Building configuration...
[OK]
R3#
```

ASW1 Authenticated NTP Client Configuration

```
Password:

ASW1>en
Password:
ASW1#configure terminal
Enter configuration commands, one per line.  End with CNTL/Z.
ASW1(config)#ntp authentication-key 1 md5 Thuthuka
ASW1(config)#ntp authenticate
ASW1(config)#ntp trusted-key 1
ASW1(config)#ntp server 10.255.0.14 key 1
ASW1(config)#end
ASW1#
*Mar 23, 04:55:39.5555: SYS-5-CONFIG_I: Configured from console by console
ASW1#
```

ASW2 Authenticated NTP Client Configuration

```
ASW2>en
Password:
ASW2#configure terminal
Enter configuration commands, one per line.  End with CNTL/Z.
ASW2(config)#ntp authentication-key 1 md5 Thuthuka
ASW2(config)#ntp authenticate
ASW2(config)#ntp trusted-key 1
ASW2(config)#ntp server 10.255.0.14 key 1
ASW2(config)#end
ASW2#
*Mar 23, 05:05:34.055: SYS-5-CONFIG_I: Configured from console by console
ASW2#
```

ASW3 Authenticated NTP Client Configuration

```
ASW3>en
Password:
ASW3#configure terminal
Enter configuration commands, one per line.  End with CNTL/Z.
ASW3(config)#ntp authentication-key 1 md5 Thuthuka
ASW3(config)#ntp authenticate
ASW3(config)#ntp trusted-key 1
ASW3(config)#ntp server 10.255.0.14 key 1
ASW3(config)#end
ASW3#
*Mar 23, 05:09:10.099: SYS-5-CONFIG_I: Configured from console by console
ASW3#wr
Building configuration...
[OK]
ASW3#
```

ASW4 Authenticated NTP Client Configuration

```
Password:

ASW4>en
Password:
ASW4#en
ASW4#configure terminal
Enter configuration commands, one per line.  End with CNTL/Z.
ASW4(config)#ntp authentication-key 1 md5 Thuthuka
ASW4(config)#ntp authenticate
ASW4(config)#ntp trusted-key 1
ASW4(config)#ntp server 10.255.0.14 key 1
ASW4(config)#end
ASW4#
*Mar 23, 05:12:13.1212: SYS-5-CONFIG_I: Configured from console by console
ASW4#wr
Building configuration...
[OK]
ASW4#
```

Explain in your report why an unauthenticated NTP deployment undermines the evidentiary value of the syslog data you have just centralised.

Sandton R1 was configured as the authoritative NTP server, with MD5 authentication and shared key. All routers and switches successfully synchronised their clocks with R1. This ensures consistent syslog timestamps, improving event correlation and forensic investigations. Without authenticated NTP, attacker could manipulate time information, making security logs unreliable. (Galleon Systems, 2015)

3.5 POPIA Control Mapping

Section 19 of the Protection of personal Information Act (POPIA) requires organisations to use suitable technical and organisational security measures to safeguard personal information against unauthorised access, loss or misuse. (Law, 2019)

Technical Control	How it would support POPIA Section 19
VLAN Segmentation	Divides the hospital network into separate sections, limiting access to sensitive patient information. (netalit, 2024)
Access control lists	Control communication between network segments to prevent unauthorised access to confidential data. (Buckman, 2026)
RADIUS AAA authentication	Verifies administrator identities and helps ensure that only authorised users can manage network devices. (Cloudi-fi.com, 2026)
SSH remote access	Protect administrator login details and management traffic through encrypted connections. (www.ssh.com, n.d.)
Centralised Syslog	Collects security events from network devices, helping administrators detect suspicious activities and investigate incidents. (NHI Mgmt Group, 2026)
Authenticated NTP	Keeps network devices clocks synchronised, ensuring accurate timestamps for security monitoring and investigating (Galleon Systems, 2015)
HSRP redundancy	Provides backup gateway functionality to maintain network availability when a primary gateway fails. (Networklessons.com, 2026)

why a network control alone is not sufficient to satisfy it.

Network security measures such as VLANs and ACLs can limit unauthorised access, but they cannot completely prevent employees with legitimate access from misusing patient information. The hospital must also introduce staff awareness training, security policies, regular risk assessment and incident respond procedures to meet POPIA requirements. (netalit, 2024) (Buckman, 2026)

Question 4

4.1 Dynamic Routing Design

Configure OSPF on R1

```
R1>en
Password:
R1#
R1#configure terminal
Enter configuration commands, one per line.  End with CNTL/Z.
R1(config)#router ospf 1
R1(config-router)#router-id 1.1.1.1
R1(config-router)#passive-interface default
R1(config-router)#no passive-interface Serial0/3/0
R1(config-router)#no passive-interface Serial0/3/1
R1(config-router)#network 10.255.0.0 0.0.0.3 area 0
R1(config-router)#network 10.255.0.4 0.0.0.3 area 0
R1(config-router)#network 10.255.0.12 0.0.0.3 area 0
R1(config-router)#network 10.255.0.16 0.0.0.3 area 0
R1(config-router)#end
R1#write memory
*Nov 05, 17:08:39.088: SYS-5-CONFIG_I: Configured from console by console
R1#write memory
Building configuration...
[OK]
R1#wr
Building configuration...
[OK]
R1#
```

Configure OSPF on R2

```
R2>en
Password:
R2#configure terminal
Enter configuration commands, one per line.  End with CNTL/Z.
R2(config)#router ospf 1
R2(config-router)#router-id 2.2.2.2
R2(config-router)#passive-interface default
R2(config-router)#no passive-interface Serial0/3/0
R2(config-router)#no passive-interface Serial0/3/1
R2(config-router)#network 10.255.0.0 0.0.0.3 area 0
R2(config-router)#network 10.255.0.8 0.0.0.3 area 0
R2(config-router)#end
R2#write memory
*Nov 05, 17:10:32.1010: SYS-5-CONFIG_I: Configured from console by console
R2#write memory
Building configuration...
[OK]
R2#wr
Building configuration...
[OK]
R2#
```

Configure OSPF on R3

```
R3>en
Password:
R3#configure terminal
Enter configuration commands, one per line.  End with CNTL/Z.
R3(config)#router ospf 1
R3(config-router)#router-id 3.3.3.3
R3(config-router)#passive-interface default
R3(config-router)#no passive-interface Serial0/3/0
R3(config-router)#no passive-interface Serial0/3/1
R3(config-router)#network 10.255.0.4 0.0.0.3 area 0
R3(config-router)#network 10.255.0.8 0.0.0.3 area 0
R3(config-router)#end
R3#write memory
*Nov 05, 17:12:16.1212: SYS-5-CONFIG_I: Configured from console by console
R3#write memory
Building configuration...
[OK]
R3#wr
Building configuration...
[OK]
R3#
```

OSPF Passive-Interface Verification on R1. The output confirms that non-adjacency interfaces are passive, while the serial WAN interfaces remain active for OSPF neighbour formation.

```
[OK]
R1#show ip protocols

Routing Protocol is "ospf 1"
  Outgoing update filter list for all interfaces is not set
  Incoming update filter list for all interfaces is not set
  Router ID 1.1.1.1
  It is an autonomous system boundary router
  Redistributing External Routes from,
  Number of areas in this router is 1. 1 normal 0 stub 0 nssa
  Maximum path: 4
  Routing for Networks:
    10.255.0.0 0.0.0.3 area 0
    10.255.0.4 0.0.0.3 area 0
    10.255.0.12 0.0.0.3 area 0
    10.255.0.16 0.0.0.3 area 0
  Passive Interface(s):
    Vlan1
    GigabitEthernet0/0
    GigabitEthernet0/1
    GigabitEthernet0/2
  Routing Information Sources:
    Gateway         Distance      Last Update
    1.1.1.1              110      00:10:19
    2.2.2.2              110      00:13:22
    3.3.3.3              110      00:13:22
  Distance: (default is 110)
```

State whether you have used a single OSPF area or multiple areas and give one reason for your choice.

Single OSPF Area 0 was used for the Sandton, Polokwane and Rustenburg sites because the network is relatively small. Using one area simplifies routing configuration, management and troubleshooting while allowing all three routers to exchange information efficiently.

OSPF WAN Redundancy on R3. The primary link was assigned a cost of 10, and the backup link a cost of 100 to prioritise the primary route.

```
R3#enable
R3#configure terminal
Enter configuration commands, one per line.  End with CNTL/Z.
R3(config)#
R3(config)#interface Serial0/3/0
R3(config-if)#ip ospf cost 10
R3(config-if)#exit
R3(config)#
R3(config)#interface Serial0/3/1
R3(config-if)#ip ospf cost 100
R3(config-if)#exit
R3(config)#
R3(config)#end
R3#write memory
*Nov 05, 17:33:39.3333: SYS-5-CONFIG_I: Configured from console by console
R3#write memory
Building configuration...
[OK]
R3#
```

OSPF WAN Redundancy on R1. The primary link was assigned a cost of 10, and the backup link a cost of 100 to prioritise the primary route.

```
R1#
R1#enable
R1#configure terminal
Enter configuration commands, one per line.  End with CNTL/Z.
R1(config)#interface Serial0/3/1
R1(config-if)#ip ospf cost 10
R1(config-if)#exit
R1(config)#interface Serial0/3/0
R1(config-if)#ip ospf cost 100
R1(config-if)#exit
R1(config)#end
R1#write memory
*Nov 05, 17:36:31.3636: SYS-5-CONFIG_I: Configured from console by console
R1#write memory
Building configuration...
[OK]
R1#
```

The screenshot shows 100 ICMP echo requests sent from Rustenburg (R3) to Sandton (10.255.0.14), with a 100% success rate and no packet loss. This confirms that the primary WAN connection was operational and stable before testing automatic failover to the backup route.

```
R3#ping
Protocol [ip]:
Target IP address: 10.255.0.14
Repeat count [5]: 100
Datagram size [100]:
Timeout in seconds [2]: 1
Extended commands [n]:
Sweep range of sizes [n]:
Type escape sequence to abort.
Sending 100, 100-byte ICMP Echos to 10.255.0.14, timeout is 1 seconds:
!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!
!!!!!!!!!!!!!!!!!!!!!!!!!!!!
Success rate is 100 percent (100/100), round-trip min/avg/max = 1/3/41 ms

R3#
```

The primary link was disabled to simulate a failure. OSPF automatically redirected traffic through the backup route, with 995 out of 1,000 pings successful, demonstrating network redundancy.

```
*Nov 05, 17:47:38.4747: SYS-5-CONFIG_I: Configured from console by console
R3#ping
Protocol [ip]:
Target IP address: 10.255.0.14
Repeat count [5]: 1000
Datagram size [100]:
Timeout in seconds [2]: 1
Extended commands [n]:
Sweep range of sizes [n]:
Type escape sequence to abort.
Sending 1000, 100-byte ICMP Echos to 10.255.0.14, timeout is 1 seconds:
!!!!!!!!!!!!!!!!!!
R3#
R3#configure terminal
Enter configuration commands, one per line.  End with CNTL/Z.
R3(config)#interface Serial0/3/0
R3(config-if)#!!!!!!!!!!!!!!!!!!!!!
R3(config-if)#exit
R3(config)#interface Serial0/3/0
R3(config-if)#!!!!!!!!!!!!shutdown
R3(config-if)#
*Nov 05, 17:48:07.4848: %LINK-5-CHANGED: Interface Serial0/3/0, changed state to administratively down
*Nov 05, 17:48:07.4848: %LINEPROTO-5-UPDOWN: Line protocol on Interface Serial0/3/0, changed state to down
*Nov 05, 17:48:07.4848: 17:48:07: %OSPF-5-ADJCHG: Process 1, Nbr 1.1.1.1 on Serial0/3/0 from FULL to DOWN, Neighbor Down: Interface down or detached
R3(config-
if)#.....!!!!!!!!!!!!!!!!!!!
```

The screenshot shows the primary Rustenburg Sandton link failing at 18:05:29. OSPF automatically installed the backup route through Polokwane (10.255.0.9), with an estimated convergence time of approximately 6.2 seconds.

```
R3#configure terminal
Enter configuration commands, one per line.  End with CNTL/Z.
R3(config)#interface Serial0/3/0
R3(config-if)#shutdown

R3(config-if)#end
*Nov 05, 18:05:29.055: %LINK-5-CHANGED: Interface Serial0/3/0, changed state to
administratively down
*Nov 05, 18:05:29.055: %LINEPROTO-5-UPDOWN: Line protocol on Interface
Serial0/3/0, changed state to down
*Nov 05, 18:05:29.055: 18:05:29: %OSPF-5-ADJCHG: Process 1, Nbr 1.1.1.1 on
Serial0/3/0 from FULL to DOWN, Neighbor Down: Interface down or detached
R3(config-if)#end
R3#
*Nov 05, 18:05:35.055: SYS-5-CONFIG_I: Configured from console by console
R3#show ip route 10.255.0.14
Routing entry for 10.255.0.12/30
Known via "ospf 1", distance 110, metric 165, type intra area
  Last update from 10.255.0.9 on Serial0/3/1, 00:00:06 ago
  Routing Descriptor Blocks:
  * 10.255.0.9, from 1.1.1.1, 00:00:06 ago, via Serial0/3/1
      Route metric is 165, traffic share count is 1

R3#show clock
18:5:41.297 UTC Thu Nov 5 2026
R3#
```

State the convergence time you measured and comment on whether it is acceptable for the traffic types the clinic carries.

The WAN failover test confirmed that OSPF automatically switched traffic from the primary Rustenburg Sandton link to the backup route through Polokwane 10.255.0.9 .The test recorded 995 successful pings out of 1000, with an estimated convergence time of 6.2 seconds. This interruption is generally acceptable for administrative traffic but may affect real time clinical monitoring and voice communications, which requires faster recovery.

Shows PAT configured using ACL 10 and overload, allowing internal hospital devices to share one public IP address.

```
R1#configure terminal
Enter configuration commands, one per line.  End with CNTL/Z.
R1(config)#
R1(config)#interface GigabitEthernet0/0
R1(config-if)#ip nat inside
R1(config-if)#exit
R1(config)#
R1(config)#interface GigabitEthernet0/1
R1(config-if)#ip nat inside
R1(config-if)#exit
R1(config)#
R1(config)#interface Serial0/3/0
R1(config-if)#ip nat inside
R1(config-if)#exit
R1(config)#
R1(config)#interface Serial0/3/1
R1(config-if)#ip nat inside
R1(config-if)#exit
R1(config)#
R1(config)#interface GigabitEthernet0/2
R1(config-if)#ip nat outside
R1(config-if)#exit
R1(config)#
R1(config)#ip nat inside source list 10 interface GigabitEthernet0/2 overload
R1(config)#
R1(config)#end
R1#write memory
*Nov 05, 18:24:41.2424: SYS-5-CONFIG_I: Configured from console by console
R1#write memory
Building configuration...
```

Shows the internal interfaces configured with ip nat inside and the ISP-facing interface GigabitEthernet0/2 configured with ip nat outside. This establishes the inside and outside interfaces required for PAT on the Sandton edge router.

```
!
interface GigabitEthernet0/0
 ip address 10.255.0.14 255.255.255.252
 ip nat inside
 duplex auto
 speed auto
!
interface GigabitEthernet0/1
 ip address 10.255.0.18 255.255.255.252
 ip nat inside
 duplex auto
 speed auto
!
interface GigabitEthernet0/2
 description CONNECTION-TO-ISP
 ip address 203.0.113.2 255.255.255.252
 ip nat outside
 duplex auto
 speed auto
!
interface Serial0/3/0
 description WAN-TO-POLOKWANE
 ip address 10.255.0.1 255.255.255.252
 ip ospf cost 100
 ip nat inside
!
interface Serial0/3/1
 description WAN-TO-RUSTENBURG
 ip address 10.255.0.5 255.255.255.252
 ip ospf cost 10
 ip nat inside
```

shows ACL 10 permitting the hospital's internal IP address range (10.10.0.0/16) for PAT on R1

```
R1#
R1#show access-lists 10
Standard IP access list 10
    permit 10.10.0.0 0.0.255.255 (8 match(es))
```

shows a successful ping from the patient Wi-Fi device to the simulated internet server (198.51.100.10), with 4 packets sent, 4 received and 0% packet loss. This confirms that the patient Wi-Fi network can reach the external server.

```
C:\>ping 198.51.100.10

Pinging 198.51.100.10 with 32 bytes of data:

Reply from 198.51.100.10: bytes=32 time=35ms TTL=125
Reply from 198.51.100.10: bytes=32 time=29ms TTL=125
Reply from 198.51.100.10: bytes=32 time=9ms TTL=125
Reply from 198.51.100.10: bytes=32 time=11ms TTL=125

Ping statistics for 198.51.100.10:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 9ms, Maximum = 35ms, Average = 21ms

C:\>
```

confirms that PAT translates the patient Wi-Fi device's private IP address (10.10.4.33) to the shared external IP address (203.0.113.2), allowing communication with the simulated internet server (198.51.100.10).

```
R1#
R1#show ip nat translations
Pro  Inside global     Inside local       Outside local      Outside global
icmp 203.0.113.2:17    10.10.4.33:17      198.51.100.10:17   198.51.100.10:17
icmp 203.0.113.2:18    10.10.4.33:18      198.51.100.10:18   198.51.100.10:18
icmp 203.0.113.2:19    10.10.4.33:19      198.51.100.10:19   198.51.100.10:19
icmp 203.0.113.2:20    10.10.4.33:20      198.51.100.10:20   198.51.100.10:20

R1#
```

shows that the patient Wi-Fi device cannot reach the internal hospital server (10.10.2.10), with packet loss.

```
Pinging 10.10.2.10 with 32 bytes of data:

Reply from 10.10.4.3: Destination host unreachable.
Reply from 10.10.4.3: Destination host unreachable.
Reply from 10.10.4.3: Destination host unreachable.
Reply from 10.10.4.3: Destination host unreachable.

Ping statistics for 10.10.2.10:
    Packets: Sent = 4, Received = 0, Lost = 4 (100% loss),

C:\>
```

4.4 Service Assurance

To meet R4, the hospital network will use Quality of service to classify and prioritise traffic according to its importance. Telemedicine voice traffic will be marked EF (46), video traffic AF41(34), and clinical workstation traffic AF21(18), while radiology image transfers will receive AF11(10) and general network traffic will use best effort (0). The network will use Class-Based Weighted Fair Queuing (CBWFQ) with Low Latency Queuing (LLQ) to give real-time telemedicine traffic priority and reserve bandwidth for clinical workstations. This approach is suitable because it reduces delays for important medical services and maintains

their responsiveness even when large radiology images are being transferred across the network. (Fortinet, 2023) (Swami, 2025) (ManageEngine Blog, 2011) (https://www.globalknowledge.com/en-ca/resources/articles/quality-of-service-part-12-low-latency-queuing, 2024)

Question 5

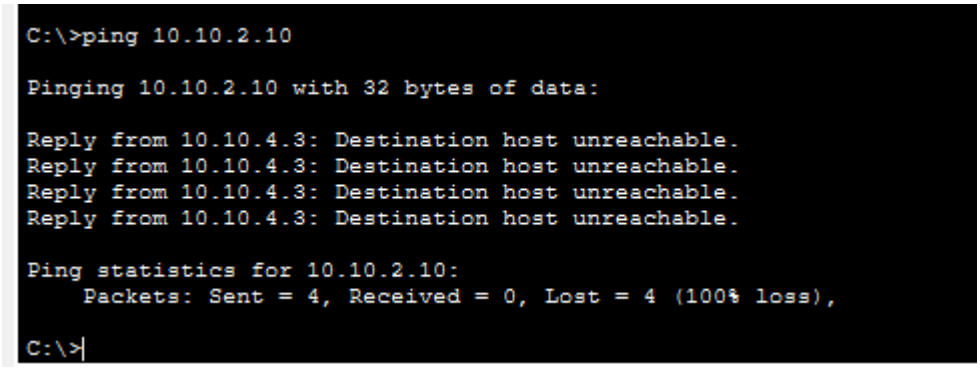
5.1 Test Plan and Evidence

R1- Medical Device Isolation Test, the medical device isolation policy was tested on MLS2 using the commands show access-lists MEDICAL-DEVICE-ISOLATION and show ip interface vlan 60. The test was expected to confirm that the ACL allowed communication with the clinical monitoring server (10.10.2.11), prevented medical devices from accessing other networks, and was applied inbound on VLAN 60. However, the test was unsuccessful because the ACL contained the correct permit and deny statements, but the VLAN 60 interface showed “Inbound access list is not set.” This means that the ACL was not active on the interface and could not enforce the required traffic restrictions through MLS2.

```
Password:
MLS2#show access-lists MEDICAL-DEVICE-ISOLATION
Extended IP access list MEDICAL-DEVICE-ISOLATION
    permit ip 10.10.1.192 0.0.0.63 host 10.10.2.11
    deny ip 10.10.1.192 0.0.0.63 any

MLS2#show ip interface vlan 60
Vlan60 is up, line protocol is up
  Internet address is 10.10.1.194/26
  Broadcast address is 255.255.255.255
  Address determined by setup command
  MTU is 1500 bytes
  Helper address is not set
  Directed broadcast forwarding is disabled
  Outgoing access list is not set
  Inbound  access list is not set
```

R2 -Patient Wi-Fi Isolation Test**,** Patient Wi-Fi isolation was tested by sending a ping from a patient wireless device to the internal hospital server. The expected result was for the connection to be blocked while internet access remained available. The test returned four unsuccessful packets with 100% packet loss and a “Destination host unreachable” message from 10.10.4.3. This confirmed that the patient device could not reach the internal server during the test. However, further verification is required to determine whether the failure was caused by the isolation ACL or a routing issue.


C:\>ping 10.10.2.10
Pinging 10.10.2.10 with 32 bytes of data:
Reply from 10.10.4.3: Destination host unreachable.
Reply from 10.10.4.3: Destination host unreachable.
Reply from 10.10.4.3: Destination host unreachable.
Reply from 10.10.4.3: Destination host unreachable.
Ping statistics for 10.10.2.10:
Packets: Sent = 4, Received = 0, Lost = 4 (100% loss),
C:\>

R3- Network Redundancy Test, Network redundancy was examined on MLS1 using the commands show standby brief, show EtherChannel summary, and show interfaces trunk. The expected outcome was for HSRP to provide active and standby gateway roles, EtherChannel to maintain bundled connections, and trunk links to carry the required VLANs. The results showed that MLS1 was active for VLANs 10, 20, 30 and 50, while operating as standby for VLANs 60, 70, 80 and 90. EtherChannel Po1 was operational with two bundled interfaces using LACP, and the trunk ports were active with native VLAN 999. These results confirmed that the redundancy configurations were operational on MLS1.

```
MLS1#show standby brief
                     P indicates configured to preempt.
                     |
Interface   Grp  Pri P State    Active          Standby         Virtual IP
Vl10        10   110 P Active   local           10.10.2.67      10.10.2.65
Vl20        20   110 P Active   local           10.10.1.3       10.10.1.1
Vl30        30   110 P Active   local           10.10.0.3       10.10.0.1
Vl50        50   110 P Active   local           10.10.1.131     10.10.1.129
Vl60        60   100 P Standby  10.10.1.194     local           10.10.1.193
Vl70        70   100 P Standby  10.10.4.3       local           10.10.4.1
Vl80        80   100 P Standby  10.10.6.3       local           10.10.6.1
Vl90        90   100 P Standby  10.10.2.3       local           10.10.2.1
MLS1#show etherchannel summary
Flags:  D - down        P - in port-channel
        I - stand-alone s - suspended
        H - Hot-standby (LACP only)
        R - Layer3      S - Layer2
        U - in use      f - failed to allocate aggregator
        u - unsuitable for bundling
        w - waiting to be aggregated
        d - default port


Number of channel-groups in use: 1
Number of aggregators:           1

Group  Port-channel  Protocol    Ports
------+-------------+-----------+-----------------------------------------------

1      Po1(SU)           LACP   Fa0/4(P) Fa0/5(P)
MLS1#show interfaces trunk
Port        Mode         Encapsulation  Status        Native vlan
Po1         on           802.1q         trunking      999
Fa0/1       on           802.1q         trunking      999
Fa0/2       on           802.1q         trunking      999

Port        Vlans allowed on trunk
Po1         10,20,30,50,60,70,80,90,999
Fa0/1       10,30,60,70,80,999
Fa0/2       10,20,30,999
```

R5-IP Addressing and VLAN Configuration The IP addressing and VLAN setup was examined on MLS1 using the commands show ip interface brief and show vlan brief. The expected outcome was for the required VLANs to be active, their interfaces to have valid IP addresses, and the routing table to contain the necessary networks. The results showed that the configured VLAN interfaces were operational, VLANs 10 to 100 and 999 were active, and the directly connected networks appeared in the routing table.

```
MLS1#
MLS1#show ip interface brief
Interface              IP-Address      OK? Method Status                Protocol
Port-channel1          unassigned      YES unset  up                    up
FastEthernet0/1        unassigned      YES unset  up                    up
FastEthernet0/2        unassigned      YES unset  up                    up
FastEthernet0/3        unassigned      YES unset  up                    up
FastEthernet0/4        unassigned      YES unset  up                    up
FastEthernet0/5        unassigned      YES unset  up                    up
FastEthernet0/6        unassigned      YES unset  up                    up
FastEthernet0/7        unassigned      YES unset  down                  down
FastEthernet0/8        unassigned      YES unset  down                  down
FastEthernet0/9        unassigned      YES unset  down                  down
FastEthernet0/10       unassigned      YES unset  down                  down
FastEthernet0/11       unassigned      YES unset  down                  down
FastEthernet0/12       unassigned      YES unset  down                  down
FastEthernet0/13       unassigned      YES unset  down                  down
FastEthernet0/14       unassigned      YES unset  down                  down
FastEthernet0/15       unassigned      YES unset  down                  down
FastEthernet0/16       unassigned      YES unset  down                  down
FastEthernet0/17       unassigned      YES unset  down                  down
FastEthernet0/18       unassigned      YES unset  down                  down
FastEthernet0/19       unassigned      YES unset  down                  down
FastEthernet0/20       unassigned      YES unset  down                  down
FastEthernet0/21       unassigned      YES unset  down                  down
FastEthernet0/22       unassigned      YES unset  down                  down
FastEthernet0/23       unassigned      YES unset  down                  down
FastEthernet0/24       unassigned      YES unset  down                  down
GigabitEthernet0/1     10.255.0.13     YES manual up                    up
GigabitEthernet0/2     unassigned      YES unset  administratively down down
Vlan1                  unassigned      YES unset  administratively down down
Vlan10                 10.10.2.66      YES manual up                    up
Vlan20                 10.10.1.2       YES manual up                    up
Vlan30                 10.10.0.2       YES manual up                    up
Vlan50                 10.10.1.130     YES manual up                    up
Vlan60                 10.10.1.195     YES manual up                    up
Vlan70                 10.10.4.2       YES manual up                    up
Vlan80                 10.10.6.2       YES manual up                    up
Vlan90                 10.10.2.2       YES manual up                    up
```

```
MLS1#show vlan brief

VLAN Name                             Status    Ports
---- -------------------------------- --------- -------------------------------
1    default                          active    Fa0/7, Fa0/8, Fa0/9, Fa0/10
                                                Fa0/11, Fa0/12, Fa0/13, Fa0/14
                                                Fa0/15, Fa0/16, Fa0/17, Fa0/18
                                                Fa0/19, Fa0/20, Fa0/21, Fa0/22
                                                Fa0/23, Fa0/24, Gig0/2
10   MANAGEMENT                       active
20   ADMIN                            active
30   CLINICAL                         active
40   MEDICAL-DEVICES                  active
50   PACS                             active
60   ICU-DEVICES                      active
70   GUEST-WIFI                       active
80   STAFF-WIFI                       active
90   SERVERS                          active    Fa0/3, Fa0/6
100  VOICE                            active
999  PARKING-NATIVE                   active
1002 fddi-default                     active
1003 token-ring-default               active
1004 fddinet-default                  active
1005 trnet-default                    active
```

R6 -OSPF Routing Test, OSPF routing was examined on R1 using the commands show ip ospf neighbor, and show ip protocols. The expected result was for the WAN routers to establish FULL neighbour relationships, exchange routing information and operate within OSPF Area 0. The test was successful for the connections examined, as R1 established FULL neighbour relationships with R2 (2.2.2.2) and R3 (3.3.3.3). The routing table also displayed an OSPF-learned route to 10.255.0.8/30, while the routing configuration confirmed Area 0. These findings demonstrate that OSPF neighbour formation and route learning are functioning on R1.

```
Password:
R1#show ip ospf neighbor


Neighbor ID     Pri   State           Dead Time   Address         Interface
3.3.3.3           0   FULL/  -        00:00:32    10.255.0.6      Serial0/3/1
2.2.2.2           0   FULL/  -        00:00:32    10.255.0.2      Serial0/3/0
R1#show ip route ospf
     10.0.0.0/8 is variably subnetted, 12 subnets, 4 masks
O       10.255.0.8 [110/110] via 10.255.0.6, 01:29:05, Serial0/3/1

R1#show ip protocols

Routing Protocol is "ospf 1"
  Outgoing update filter list for all interfaces is not set
  Incoming update filter list for all interfaces is not set
  Router ID 1.1.1.1
  Number of areas in this router is 1. 1 normal 0 stub 0 nssa
  Maximum path: 4
  Routing for Networks:
    10.255.0.0 0.0.0.3 area 0
    10.255.0.4 0.0.0.3 area 0
    10.255.0.12 0.0.0.3 area 0
    10.255.0.16 0.0.0.3 area 0
  Passive Interface(s):
    Vlan1
    GigabitEthernet0/0
    GigabitEthernet0/1
    GigabitEthernet0/2
  Routing Information Sources:
    Gateway         Distance      Last Update
    1.1.1.1              110      00:16:23
    2.2.2.2              110      00:16:23
    3.3.3.3              110      00:28:49
  Distance: (default is 110)
```

R7- WAN Failover Test, WAN redundancy was evaluated by checking the routing table and OSPF neighbour relationships on R3 using show ip route and show ip ospf neighbor. The expected result was for the router to use its primary connection during normal operation and automatically switch to the backup route through Polokwane if the primary link failed. The results confirmed that R3 had established FULL OSPF neighbour relationships with R1 and R2, while its routing table showed routes through the primary connection.

```
R3>en
Password:
R3#show ip route
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
       i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter area
       * - candidate default, U - per-user static route, o - ODR
       P - periodic downloaded static route

Gateway of last resort is not set

     10.0.0.0/8 is variably subnetted, 8 subnets, 3 masks
S       10.10.2.0/26 [1/0] via 10.255.0.5
O       10.255.0.0/30 [110/110] via 10.255.0.5, 01:20:01, Serial0/3/0
C       10.255.0.4/30 is directly connected, Serial0/3/0
L       10.255.0.6/32 is directly connected, Serial0/3/0
C       10.255.0.8/30 is directly connected, Serial0/3/1
L       10.255.0.10/32 is directly connected, Serial0/3/1
O       10.255.0.12/30 [110/11] via 10.255.0.5, 01:32:21, Serial0/3/0
O       10.255.0.16/30 [110/11] via 10.255.0.5, 01:32:21, Serial0/3/0

R3#show ip ospf neighbor


Neighbor ID     Pri   State           Dead Time   Address         Interface
2.2.2.2           0   FULL/  -        00:00:32    10.255.0.9      Serial0/3/1
1.1.1.1           0   FULL/  -        00:00:38    10.255.0.5      Serial0/3/0
R3#
```

R8- NAT/PAT was tested on R1 using the show running-config and show ip route commands to verify the NAT configuration and routing. The expected result was that internal private IP addresses would be translated to the public IP address for internet access. The observed results confirmed that NAT inside and outside interfaces were configured.

```
R1#show running-config | include ip nat
 ip nat inside
 ip nat inside
 ip nat outside
 ip nat inside
 ip nat inside
ip nat inside source list 10 interface GigabitEthernet0/2 overload
R1#show running-config | include access-list
access-list 10 permit 10.10.0.0 0.0.255.255
R1#show ip route
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
       i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter area
       * - candidate default, U - per-user static route, o - ODR
       P - periodic downloaded static route

Gateway of last resort is 203.0.113.1 to network 0.0.0.0

     10.0.0.0/8 is variably subnetted, 23 subnets, 7 masks
S       10.10.2.0/26 [1/0] via 10.255.0.13
S       10.10.2.64/26 [1/0] via 10.255.0.13
S       10.10.4.0/23 [1/0] via 10.255.0.17
O       10.10.16.0/25 [110/101] via 10.255.0.2, 00:45:00, Serial0/3/0
O       10.10.16.128/26 [110/101] via 10.255.0.2, 00:45:00, Serial0/3/0
O       10.10.16.192/27 [110/101] via 10.255.0.2, 00:45:00, Serial0/3/0
O       10.10.16.224/27 [110/101] via 10.255.0.2, 00:45:00, Serial0/3/0
```

‘

R9- VLAN and trunk configurations were tested on MLS1 using the show interfaces trunk and show vlan brief commands. The expected result was that the required VLANs would be active, trunk links would use 802.1Q encapsulation, native VLAN 999 would be configured, and only necessary VLANs would be allowed across each trunk. The observed results confirmed that VLANs 10, 20, 30, 40, 50, 60, 70, 80, 90 and 100 were active, while Port-channel1, Fa0/1 and Fa0/2 were successfully trunking with native VLAN 999. The permitted VLANs were also in the spanning-tree forwarding state. The MLS1 test passed.

```
MLS1#show interfaces trunk
Port        Mode         Encapsulation  Status        Native vlan
Po1         on           802.1q         trunking      999
Fa0/1       on           802.1q         trunking      999
Fa0/2       on           802.1q         trunking      999

Port        Vlans allowed on trunk
Po1         10,20,30,50,60,70,80,90,999
Fa0/1       10,30,60,70,80,999
Fa0/2       10,20,30,999

Port        Vlans allowed and active in management domain
Po1         10,20,30,50,60,70,80,90,999
Fa0/1       10,30,60,70,80,999
Fa0/2       10,20,30,999

Port        Vlans in spanning tree forwarding state and not pruned
Po1         10,20,30,50,60,70,80,90,999
Fa0/1       10,30,60,70,80,999
Fa0/2       10,20,30,999

MLS1#show vlan brief

VLAN Name                             Status    Ports
---- -------------------------------- --------- -------------------------------
1    default                          active    Fa0/7, Fa0/8, Fa0/9, Fa0/10
                                                Fa0/11, Fa0/12, Fa0/13, Fa0/14
                                                Fa0/15, Fa0/16, Fa0/17, Fa0/18
                                                Fa0/19, Fa0/20, Fa0/21, Fa0/22
                                                Fa0/23, Fa0/24, Gig0/2
10   MANAGEMENT                       active
20   ADMIN                            active
30   CLINICAL                         active
40   MEDICAL-DEVICES                  active
50   PACS                             active
60   ICU-DEVICES                      active
70   GUEST-WIFI                       active
80   STAFF-WIFI                       active
90   SERVERS                          active    Fa0/3, Fa0/6
100  VOICE                            active
```

R10- Access-layer security was tested on ASW1 using the show ip dhcp snooping and show ip arp inspection commands. The expected result was that DHCP snooping would prevent unauthorised DHCP servers from distributing addresses, while Dynamic ARP Inspection would protect against ARP spoofing. The observed results confirmed that both DHCP snooping and DAI were enabled on VLANs 30 and 60. DHCP snooping also showed Fa0/1 as trusted and the remaining listed interfaces as untrusted.

```
ASW1#show ip dhcp snooping
Switch DHCP snooping is enabled
DHCP snooping is configured on following VLANs:
30,60
Insertion of option 82 is disabled
Option 82 on untrusted port is not allowed
Verification of hwaddr field is enabled
Interface                  Trusted    Rate limit (pps)
-----------------------    -------    ----------------
FastEthernet0/1            yes        unlimited
FastEthernet0/4            no         unlimited
FastEthernet0/3            no         unlimited
FastEthernet0/2            no         unlimited
ASW1#show ip arp inspection

Source Mac Validation      : Disabled
Destination Mac Validation : Disabled
IP Address Validation      : Disabled

 Vlan     Configuration    Operation   ACL Match          Static ACL
 ----     -------------    ---------   ---------          ----------
   30     Enabled          Active
   60     Enabled          Active

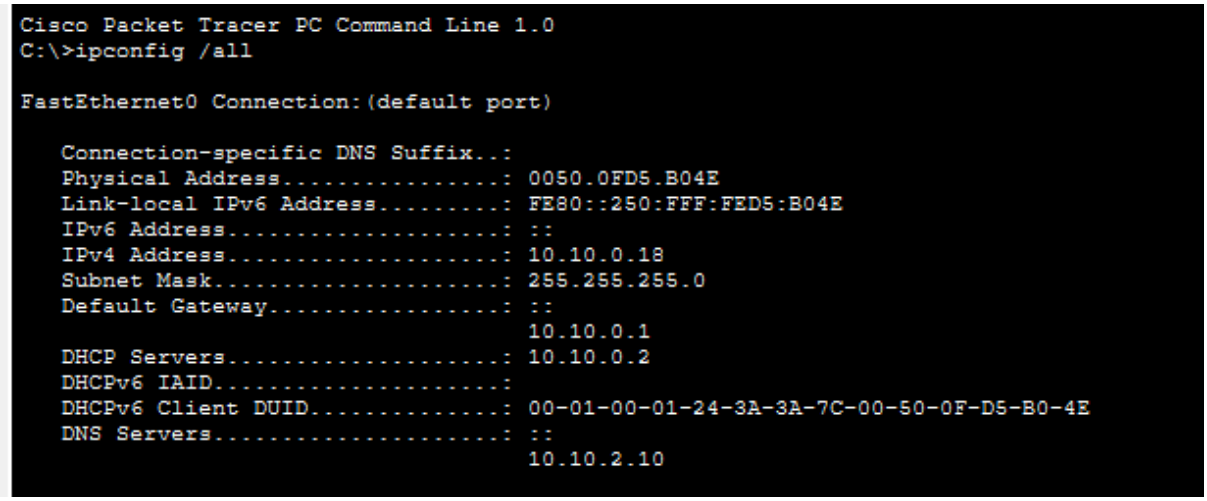
 Vlan     ACL Logging      DHCP Logging      Probe Logging
 ----     -----------      ------------      -------------
   30     Deny             Deny              Off
   60     Deny             Deny              Off

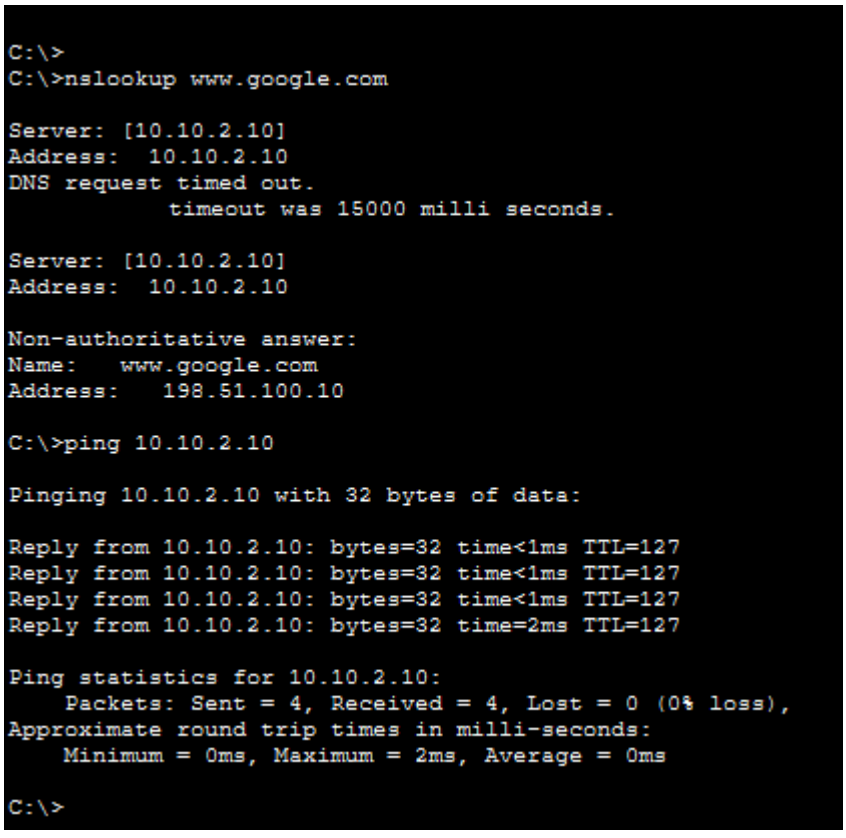
 Vlan      Forwarded        Dropped     DHCP Drops      ACL Drops
 ----      ---------        -------     ----------      ---------
   30              0              0              0              0
   60              0              0              0              0

 Vlan   DHCP Permits    ACL Permits  Probe Permits   Source MAC Failures
 ----   ------------    -----------  -------------   -------------------
   30              0              0              0                     0
   60              0              0              0                     0

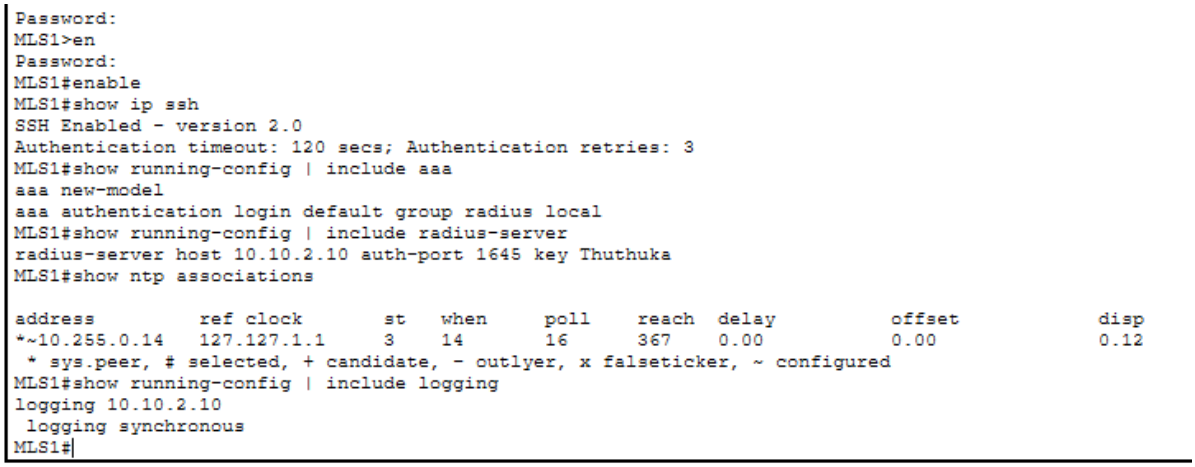
 Vlan   Dest MAC Failures   IP Validation Failures   Invalid Protocol Data
```

R11- Network services were tested on a Sandton Clinical PC using the ipconfig /all, nslookup www.google.com, and ping 10.10.2.10 commands. The expected result was that the PC would receive the correct network configuration through DHCP, successfully resolve domain names using the DNS server, and communicate with the server. The observed results confirmed that the PC had IP address 10.10.0.18, subnet mask 255.255.255.0, default gateway 10.10.0.1, DHCP server 10.10.0.2, and DNS server 10.10.2.10. The DNS lookup successfully resolved www.google.com to 198.51.100.10 , R11 passed for the tested client..





R12 - Network management and monitoring were tested on MLS1 using the show ip ssh, show running-config, and show ntp associations commands. The expected result was that SSH version 2 would provide secure remote management, AAA would support RADIUS authentication with local fallback, NTP would synchronise the device clock, and syslog would forward network events to a central server. The observed results confirmed that SSH version 2 was enabled, AAA authentication was configured with RADIUS server 10.10.2.10, and NTP was successfully synchronised with server 10.255.0.14. Remote logging was also configured to server 10.10.2.10.



```
Password:
MLS1>en
Password:
MLS1#enable
MLS1#show ip ssh
SSH Enabled - version 2.0
Authentication timeout: 120 secs; Authentication retries: 3
MLS1#show running-config | include aaa
aaa new-model
aaa authentication login default group radius local
MLS1#show running-config | include radius-server
radius-server host 10.10.2.10 auth-port 1645 key Thuthuka
MLS1#show ntp associations

address         ref clock       st   when     poll    reach  delay          offset            disp
*~10.255.0.14   127.127.1.1     3    14       16      367    0.00           0.00              0.12
 * sys.peer, # selected, + candidate, - outlyer, x falseticker, ~ configured
MLS1#show running-config | include logging
logging 10.10.2.10
 logging synchronous
MLS1#
```

5.2 Evaluation of the Design

The single points of failure that remain after your resilience measures

Although HSRP, EtherChannel and redundant WAN links improve reliability, the Sandton edge router, ISP connection and central servers remain potential single points of failure. Adding backup routers, ISP links and redundant servers would improve availability.

How your design would need to change if the Group doubled in size

If the hospital group doubles in size, additional switches, access points, VLANs and IP addresses will be needed. Network bandwidth, server capacity and DHCP services should also be expanded to support more users and medical devices.

At least one requirement that your implementation satisfies in Packet Tracer but that would need different treatment in a production deployment.

PAT works successfully in Packet Tracer, allowing internal devices to share one external IP address. However, a real hospital network would require a next-generation firewall, stronger traffic inspection and a reliable ISP connection to provide better security.

At least two security assumptions your design depends on which lie outside the network.

The design assumes that employees use strong passwords and follow security policies. It also depends on medical devices and servers being regularly updated, protected against malware and physically secured against unauthorised access.

5.3 Reflection and Troubleshooting Log

If I repeated this project, I would plan the network configuration more carefully and test each component before moving to the next question. I would verify ACL placement, IP addressing, routing and PAT earlier to identify problems quickly. I would also save configuration regularly and document troubleshooting steps to reduce errors and improve network reliability.

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